Exemple d'utilisation du package DegradSim

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Préparation

Installation du package depuis GitHub :

```
remotes::install_github("rytracetito/levy_packages", subdir = "DegradSim")
```

Les parties I et II utilisent les sept **processus de Lévy** du package (`A = NULL` par défaut). La partie III montre les **processus de Lévy à échelle de temps** (*time-scaled Lévy processes*) $X_t = Y_{A(t)}$, obtenus avec une fonction d'échelle de temps A (*time-scaling function*) : gamma, Wiener et inverse gaussien.

Paramètres utilisés :

Processus	Fonction	Paramètres
Gamma	`sim_gamma`	`a = 2`, `b = 3` (taux)
Wiener	`sim_wiener`	`mu = 1.5`, `sigma = 2`
Inverse gaussien	`sim_ig`	`mu = 0.5`, `eta = 2`
Variance-gamma	`sim_vg`	`theta = 0.5`, `sigma = 1.5`, `nu = 2`
Meixner	`sim_meixner`	`a = 1`, `b = 0.3`, `d = 1.5`, `m = 0.2`
Normal inverse gaussien	`sim_nig`	`alpha = 2`, `beta = 0.5`, `delta = 1`, `mu = 0.2`

Processus	Fonction	Paramètres
Poisson composé	`sim_cpoisson`	`lambda = 0.5`, sauts exponentiels de moyenne 1 (`rjump` par défaut)

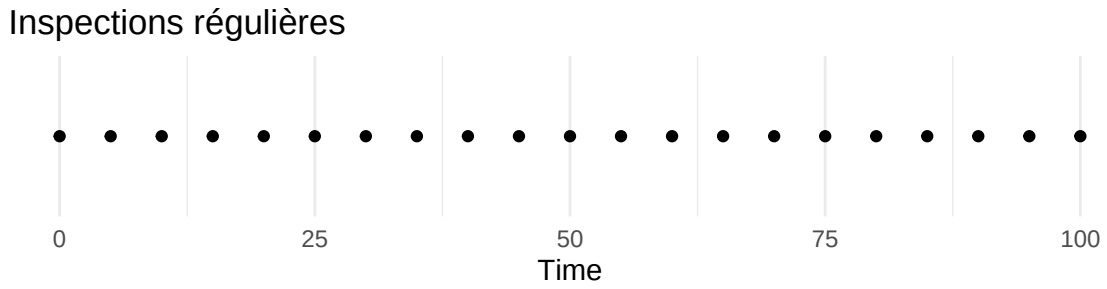
Dans chaque cas, les trajectoires simulées sont d'abord **stockées** dans un objet `x`, puis utilisées : premières valeurs, accroissements de la première trajectoire (arrondis à 3 décimales), puis graphique. Les accroissements montrent qu'un palier presque plat sur le graphique correspond à de petits accroissements, pas à des accroissements nuls (sauf pour le Poisson composé, constant entre deux chocs). Quand toutes les trajectoires ont les mêmes instants d'inspection, `x` est une matrice : une colonne `t`, puis une colonne par trajectoire (`X1`, `X2`, ...).

I - Une trajectoire

Une trajectoire inspectée 20 fois entre 0 et 100. `x` est une matrice à deux colonnes : `t` et `X1`.

Cas 1 : inspections régulières

```
t1 <- inspection_times("regular", n = 20, tmax = 100)$time
plot_plan(t1, "Inspections régulières")
```



Modèle 1 : gamma

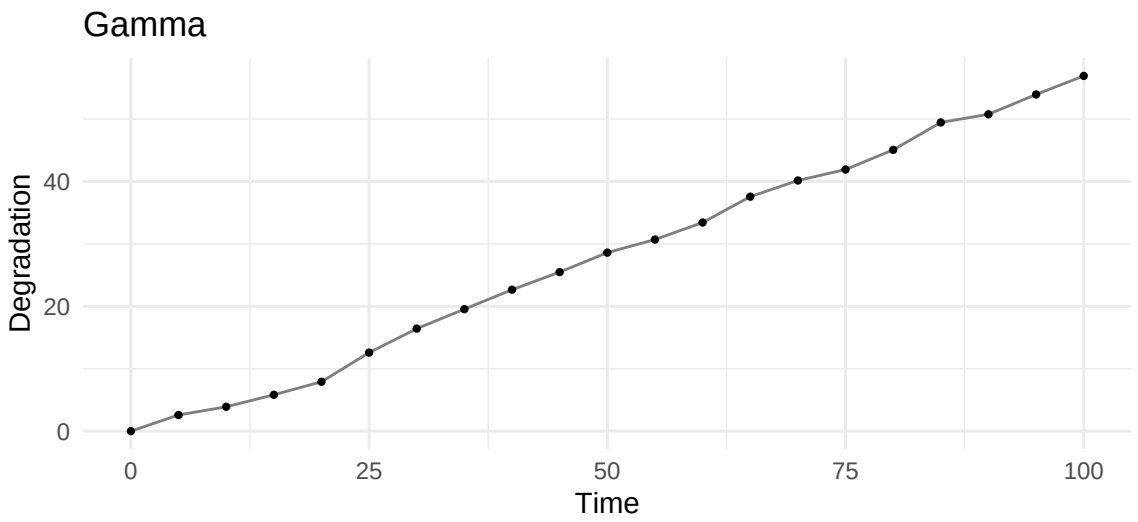
```
set.seed(11)
x <- sim_gamma(t1, a = 2, b = 3)
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  2.588550
## [3,] 10  3.900945
## [4,] 15  5.822353
## [5,] 20  7.910152
## [6,] 25 12.582689
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.589 1.312 1.921 2.088 4.673 3.841 3.120 3.116 2.819 3.123 2.095 2.725
## [13] 4.144 2.588 1.757 3.150 4.395 1.307 3.174 2.977
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

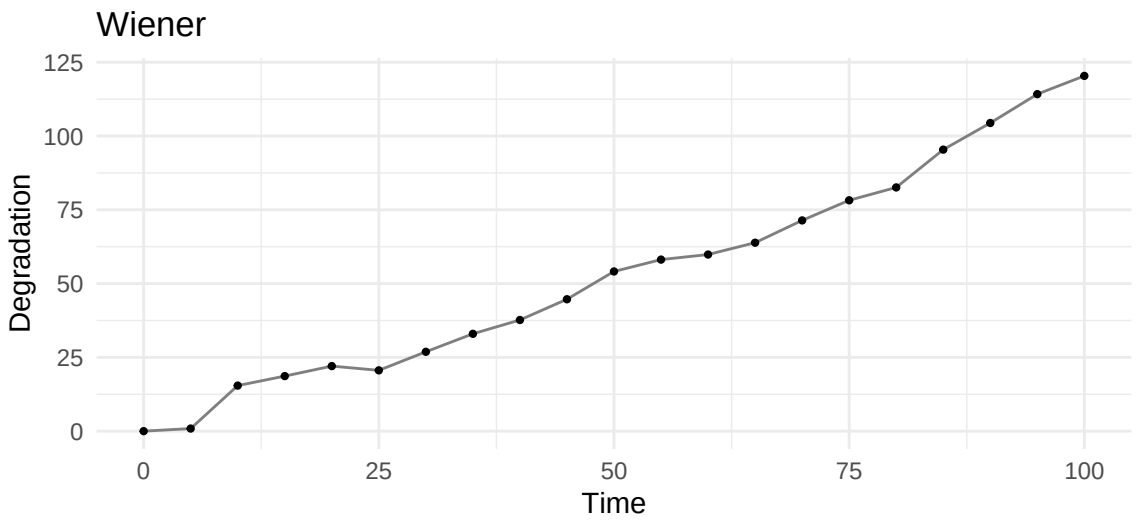
```
set.seed(12)
x <- sim_wiener(t1, mu = 1.5, sigma = 2)
head(x)
```

```
##       t         X1
## [1,]  0  0.0000000
## [2,]  5  0.8787004
## [3,] 10 15.4320167
## [4,] 15 18.6533253
## [5,] 20 22.0389368
## [6,] 25 20.6052097
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.879 14.553  3.221  3.386 -1.434  6.282  6.090  4.690  7.024  9.414
## [11]  4.022  1.714  4.014  7.553  6.818  4.354 12.817  9.023  9.767  6.188
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

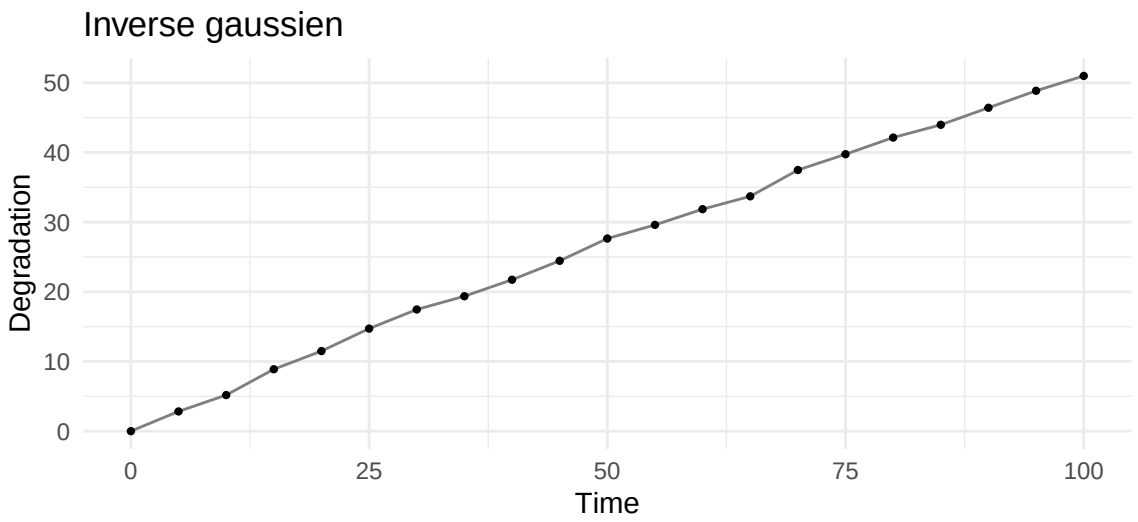
```
set.seed(13)
x <- sim_ig(t1, mu = 0.5, eta = 2)
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  2.829678
## [3,] 10  5.177833
## [4,] 15  8.886486
## [5,] 20 11.493417
## [6,] 25 14.718884
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.830 2.348 3.709 2.607 3.225 2.743 1.901 2.371 2.713 3.199 1.959 2.255
## [13] 1.846 3.775 2.266 2.394 1.832 2.444 2.437 2.137
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

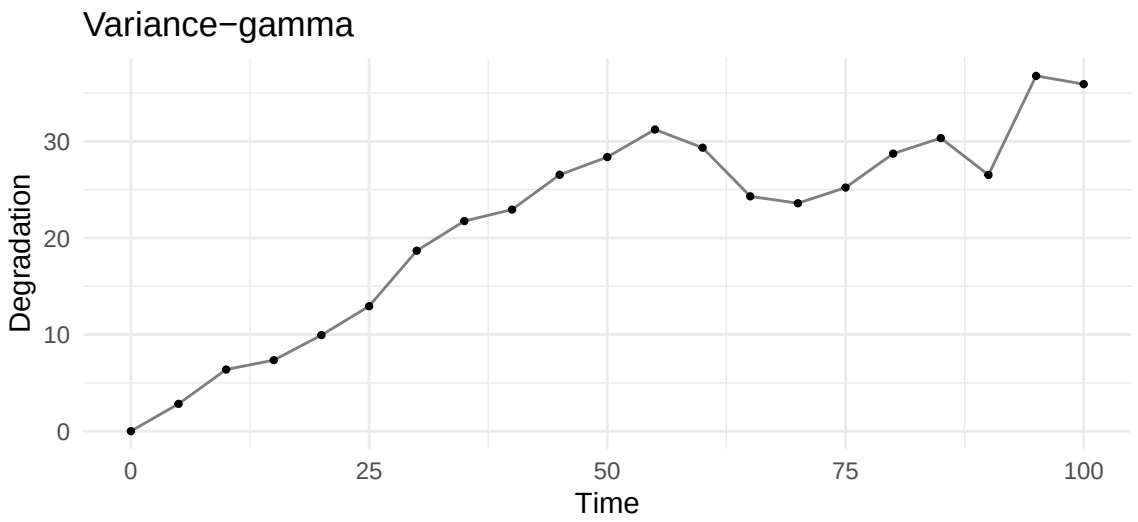
```
set.seed(14)
x <- sim_vg(t1, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  2.826550
## [3,] 10  6.375562
## [4,] 15  7.349937
## [5,] 20  9.939181
## [6,] 25 12.934933
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.827  3.549  0.974  2.589  2.996  5.745  3.070  1.187  3.601  1.838
## [11]  2.847 -1.879 -5.037 -0.715  1.625  3.514  1.609 -3.813 10.250 -0.850
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

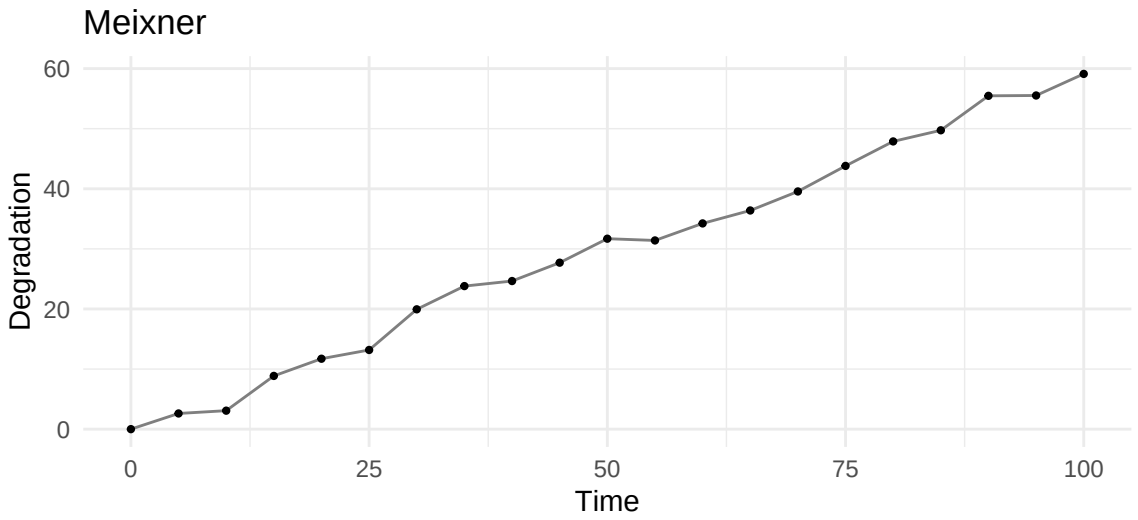
```
set.seed(15)
x <- sim_meixner(t1, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##       t        X1
## [1,]  0  0.000000
## [2,]  5  2.609114
## [3,] 10  3.073932
## [4,] 15  8.858509
## [5,] 20 11.718557
## [6,] 25 13.175516
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.609  0.465  5.785  2.860  1.457  6.765  3.865  0.841  3.056  3.990
## [11] -0.291  2.835  2.153  3.165  4.254  4.075  1.855  5.735  0.053  3.594
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

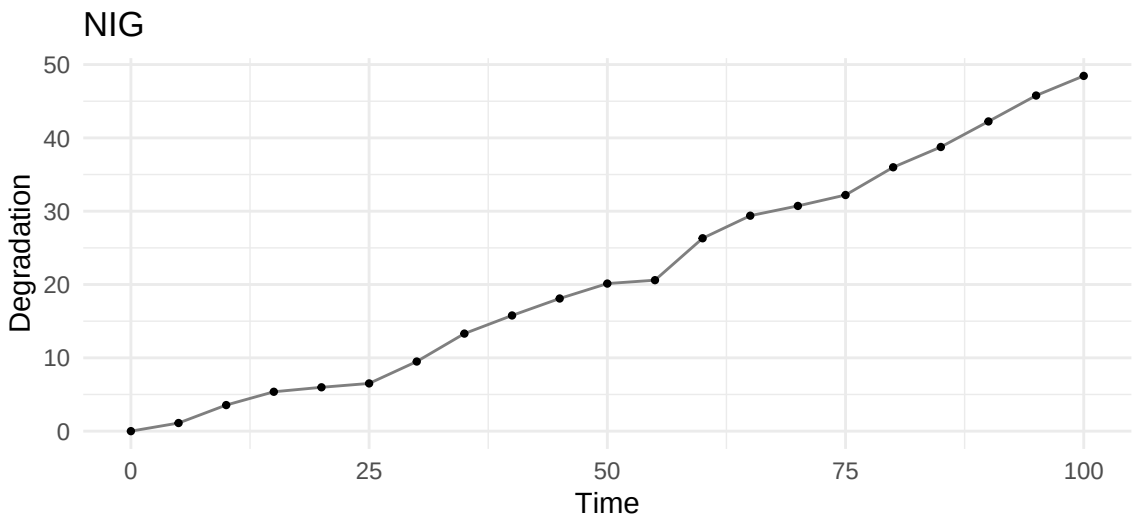
```
set.seed(16)
x <- sim_nig(t1, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##       t       X1
## [1,]  0 0.000000
## [2,]  5 1.106802
## [3,] 10 3.555632
## [4,] 15 5.369281
## [5,] 20 5.980709
## [6,] 25 6.508225
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.107 2.449 1.814 0.611 0.528 2.987 3.809 2.473 2.314 2.036 0.464 5.717
## [13] 3.078 1.331 1.489 3.784 2.767 3.484 3.534 2.677
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

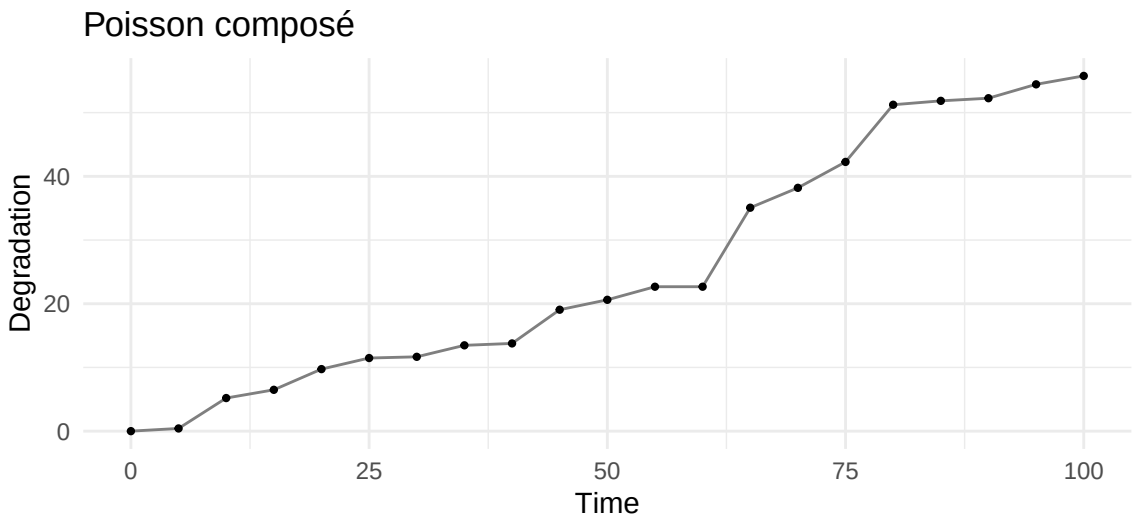
```
set.seed(17)
x <- sim_cpoisson(t1, lambda = 0.5)
head(x)
```

```
##       t         X1
## [1,]  0  0.0000000
## [2,]  5  0.4210759
## [3,] 10  5.1945031
## [4,] 15  6.4873965
## [5,] 20  9.7368579
## [6,] 25 11.4769556
```

```
round(diff(x[, "X1"]), 3)
```

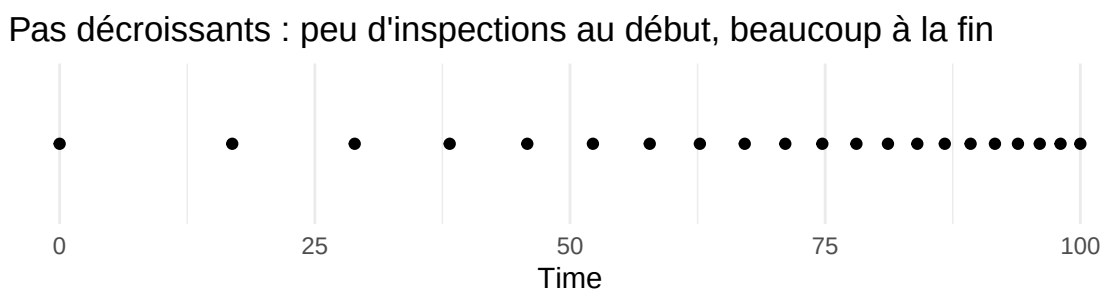
```
##  [1]  0.421  4.773  1.293  3.249  1.740  0.191  1.804  0.298  5.292  1.554
## [11]  2.055  0.000 12.406  3.120  4.068  8.975  0.614  0.413  2.177  1.325
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
0
20
40
0
25
50
75
100
Time

Cas 2 : pas décroissants

```
t2 <- inspection_times("decreasing", n = 20, tmax = 100)$time
plot_plan(t2, "Pas décroissants : peu d'inspections au début, beaucoup à la fin")
```



Modèle 1 : gamma

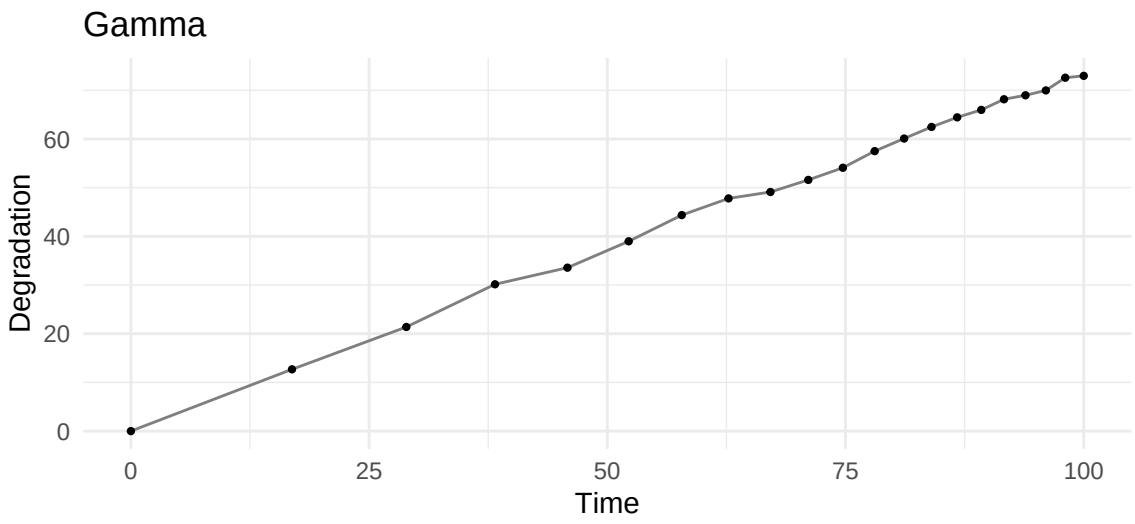
```
set.seed(21)
x <- sim_gamma(t2, a = 2, b = 3)
head(x)
```

```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 12.68436
## [3,] 28.90648 21.38241
## [4,] 38.21229 30.15090
## [5,] 45.81569 33.56269
## [6,] 52.24427 38.98223
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 12.684  8.698  8.768  3.412  5.420  5.385  3.423  1.320  2.481  2.496
## [11]  3.416  2.586  2.386  1.966  1.530  2.184  0.821  0.999  2.611  0.380
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

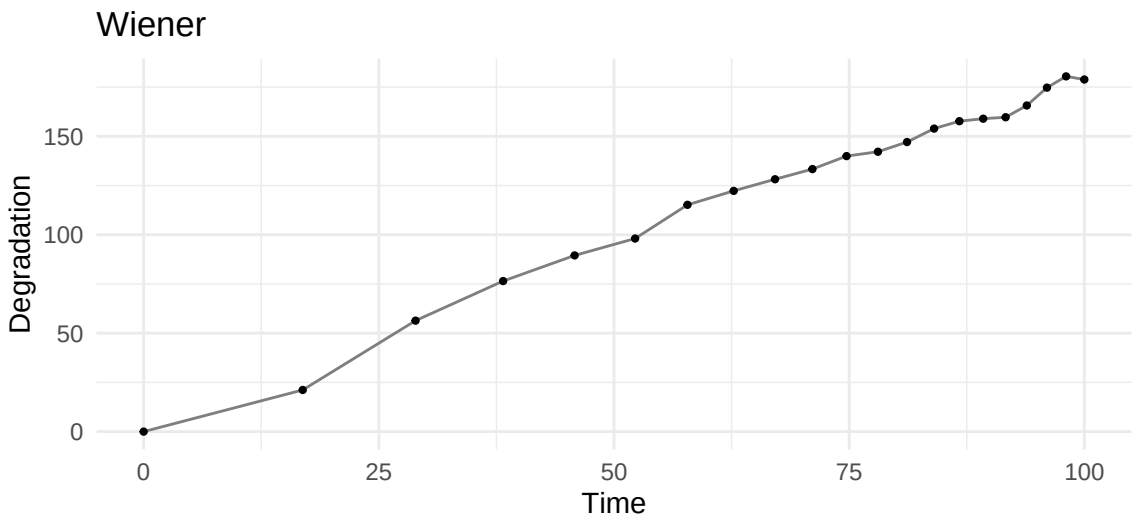
```
set.seed(22)
x <- sim_wiener(t2, mu = 1.5, sigma = 2)
head(x)
```

```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 21.15190
## [3,] 28.90648 56.36371
## [4,] 38.21229 76.47126
## [5,] 45.81569 89.49119
## [6,] 52.24427 98.07444
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 21.152 35.212 20.108 13.020  8.583 17.123  7.075  5.908  5.165  6.588
## [11]  2.216  4.924  6.837  3.762  1.271  0.728  5.970  9.068  5.724 -1.591
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

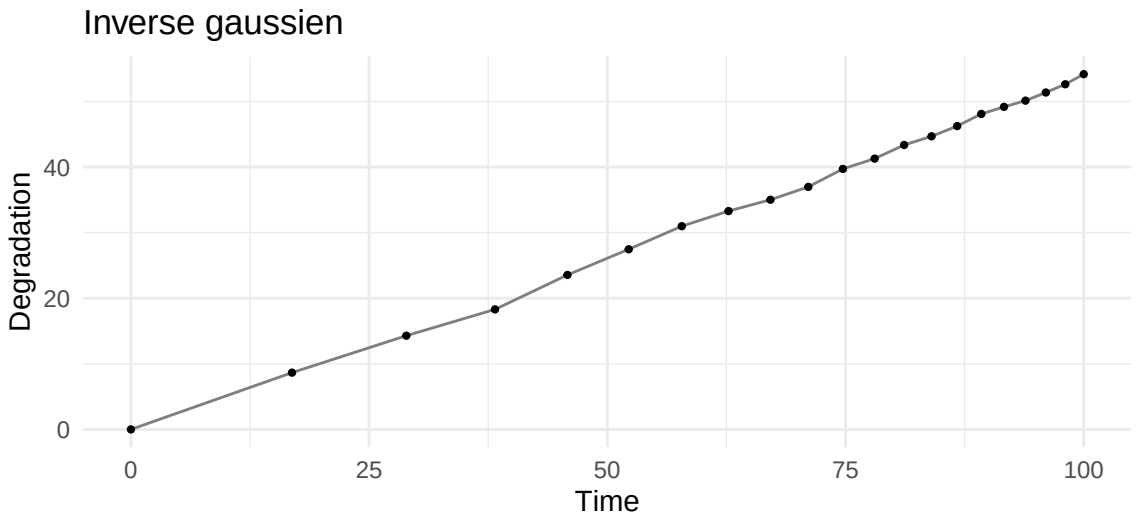
```
set.seed(23)
x <- sim_ig(t2, mu = 0.5, eta = 2)
head(x)
```

```
##             t        X1
## [1,]  0.00000  0.000000
## [2,] 16.90921  8.655577
## [3,] 28.90648 14.289435
## [4,] 38.21229 18.296030
## [5,] 45.81569 23.551265
## [6,] 52.24427 27.462389
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 8.656 5.634 4.007 5.255 3.911 3.519 2.307 1.724 1.965 2.736 1.572 2.079
## [13] 1.321 1.558 1.849 1.079 0.948 1.244 1.255 1.538
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

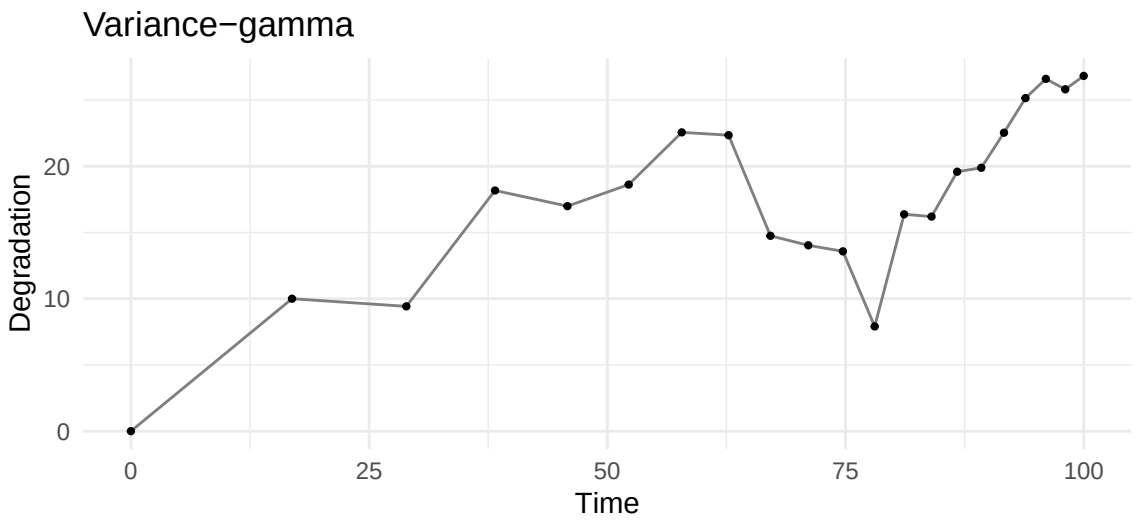
```
set.seed(24)
x <- sim_vg(t2, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 10.00188
## [3,] 28.90648  9.42331
## [4,] 38.21229 18.17461
## [5,] 45.81569 16.99351
## [6,] 52.24427 18.62047
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 10.002 -0.579  8.751 -1.181  1.627  3.944 -0.213 -7.600 -0.718 -0.455
## [11] -5.672  8.471 -0.174  3.388  0.298  2.649  2.613  1.452 -0.788  1.012
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

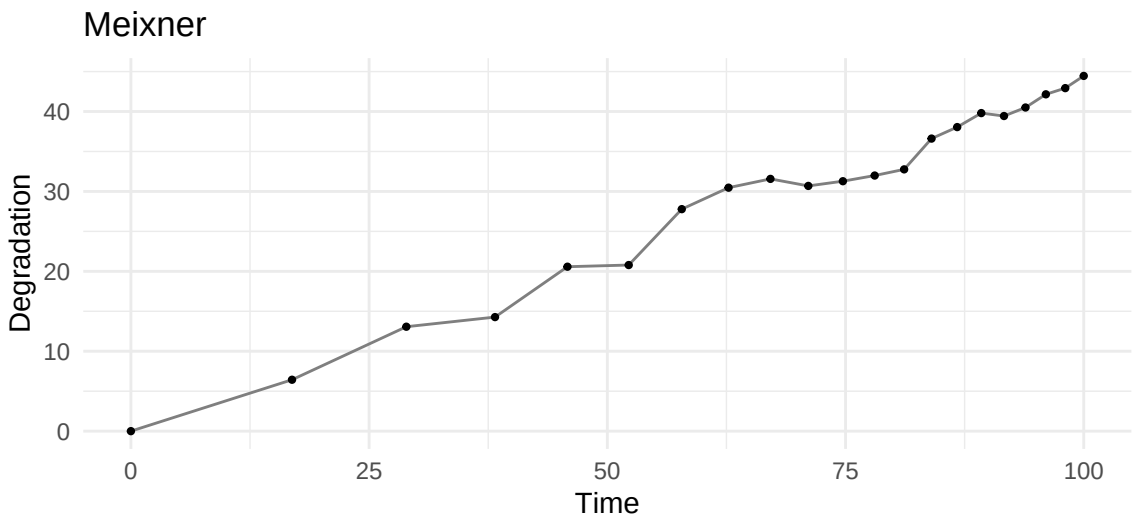
```
set.seed(25)
x <- sim_meixner(t2, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##             t        X1
## [1,]  0.00000  0.000000
## [2,] 16.90921  6.431984
## [3,] 28.90648 13.068405
## [4,] 38.21229 14.274293
## [5,] 45.81569 20.578044
## [6,] 52.24427 20.786662
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  6.432  6.636  1.206  6.304  0.209  6.996  2.688  1.098 -0.879  0.589
## [11]  0.708  0.773  3.853  1.438  1.752 -0.366  1.064  1.650  0.775  1.531
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

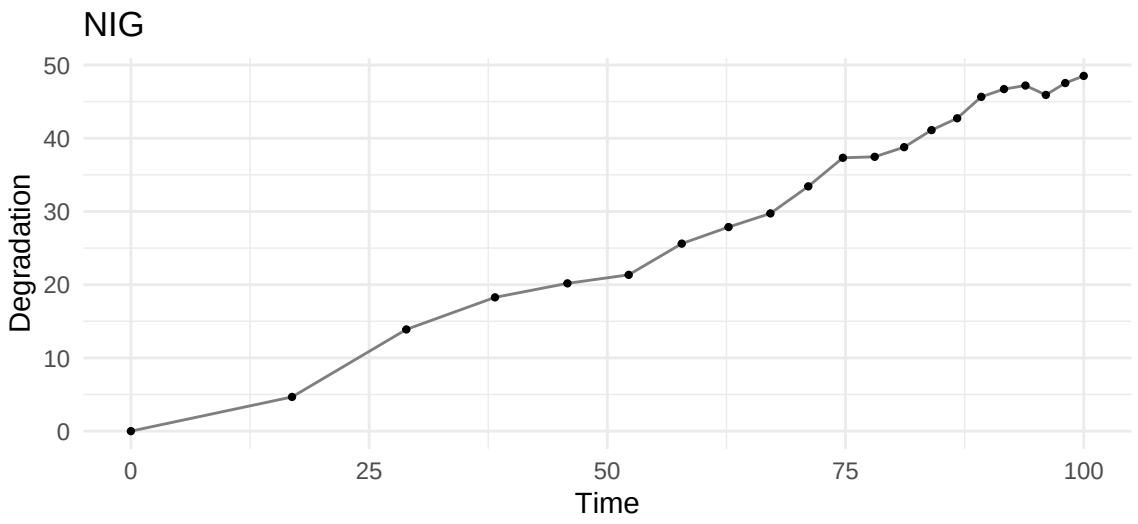
```
set.seed(26)
x <- sim_nig(t2, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##             t        X1
## [1,]  0.00000  0.000000
## [2,] 16.90921  4.668869
## [3,] 28.90648 13.887229
## [4,] 38.21229 18.270337
## [5,] 45.81569 20.180364
## [6,] 52.24427 21.346607
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  4.669  9.218  4.383  1.910  1.166  4.261  2.262  1.864  3.696  3.902
## [11]  0.132  1.316  2.332  1.614  2.929  1.057  0.483 -1.279  1.621  0.976
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

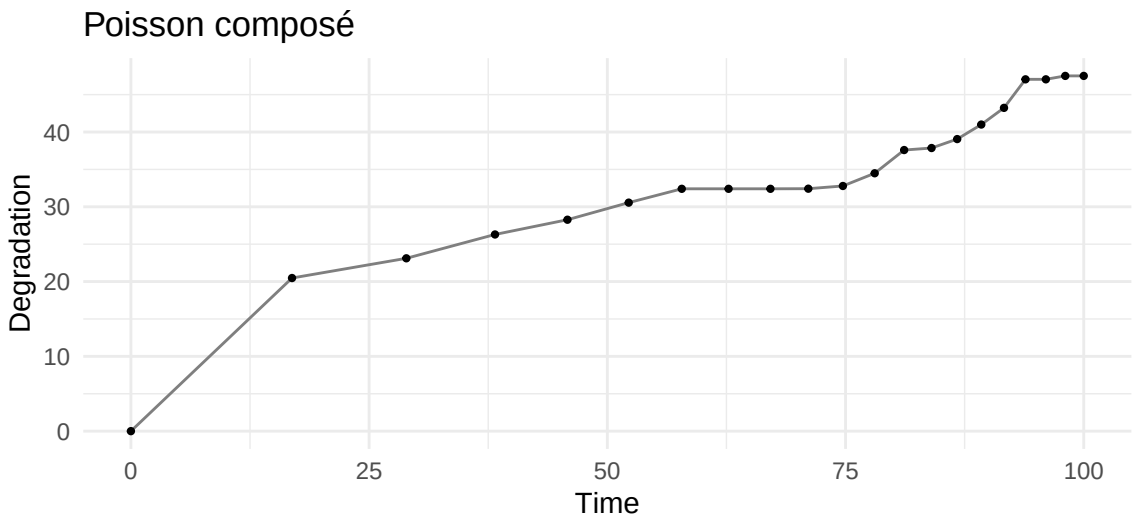
```
set.seed(27)
x <- sim_cpoisson(t2, lambda = 0.5)
head(x)
```

```
##             t       X1
## [1,]  0.00000  0.00000
## [2,] 16.90921 20.47580
## [3,] 28.90648 23.11248
## [4,] 38.21229 26.29892
## [5,] 45.81569 28.27167
## [6,] 52.24427 30.55641
```

```
round(diff(x[, "X1"]), 3)
```

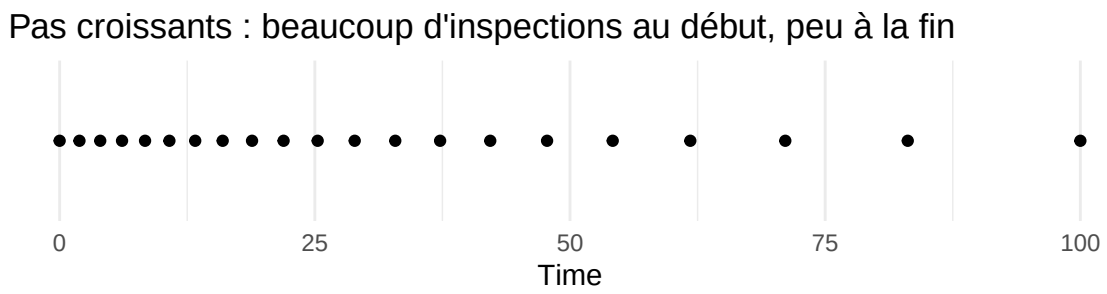
```
##  [1] 20.476  2.637  3.186  1.973  2.285  1.850  0.000  0.000  0.015  0.363
## [11]  1.700  3.106  0.269  1.196  1.943  2.232  3.814  0.000  0.459  0.000
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
0
10
20
30
40
0
25
50
75
100
Time

Cas 3 : pas croissants

```
t3 <- inspection_times("increasing", n = 20, tmax = 100)$time
plot_plan(t3, "Pas croissants : beaucoup d'inspections au début, peu à la fin")
```



Modèle 1 : gamma

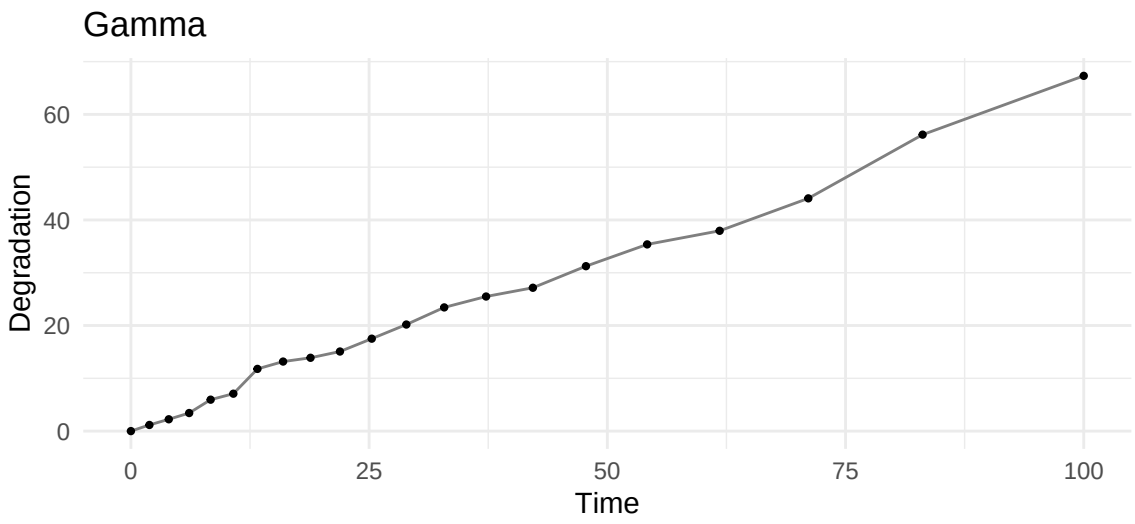
```
set.seed(31)
x <- sim_gamma(t3, a = 2, b = 3)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  1.940035 1.161003
## [3,]  3.974743 2.237610
## [4,]  6.113840 3.420305
## [5,]  8.368618 5.951271
## [6,] 10.752309 7.083982
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.161  1.077  1.183  2.531  1.133  4.699  1.389  0.714  1.189  2.432
## [11]  2.680  3.245  2.064  1.658  4.087  4.129  2.576  6.145 12.071 11.140
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

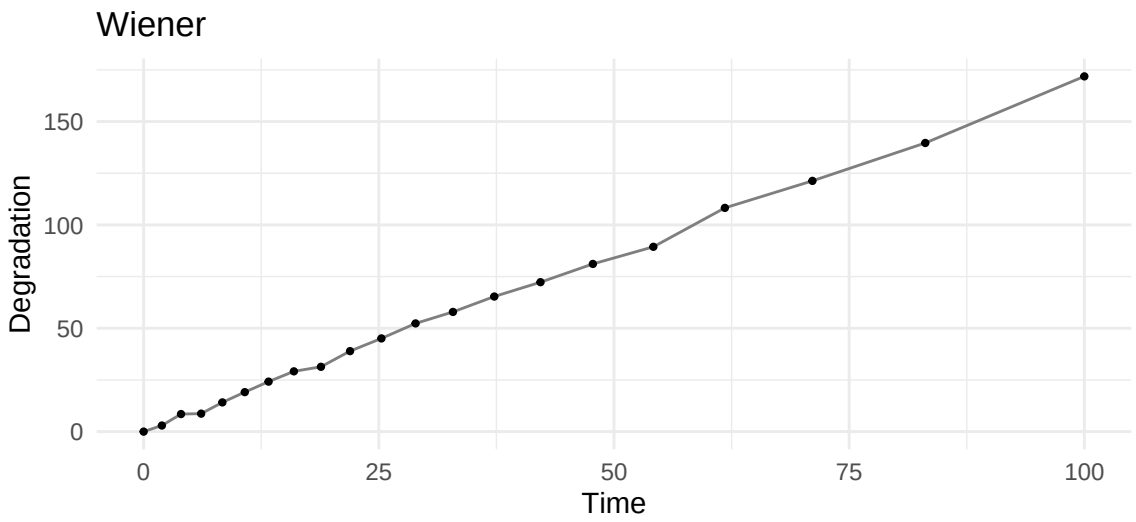
```
set.seed(32)
x <- sim_wiener(t3, mu = 1.5, sigma = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  1.940035  2.950837
## [3,]  3.974743  8.494273
## [4,]  6.113840  8.696041
## [5,]  8.368618 14.137385
## [6,] 10.752309 19.100712
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.951  5.543  0.202  5.441  4.963  5.087  4.971  2.198  7.588  6.144
## [11]  7.254  5.563  7.444  6.951  8.826  8.283 18.828 13.065 18.341 32.221
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

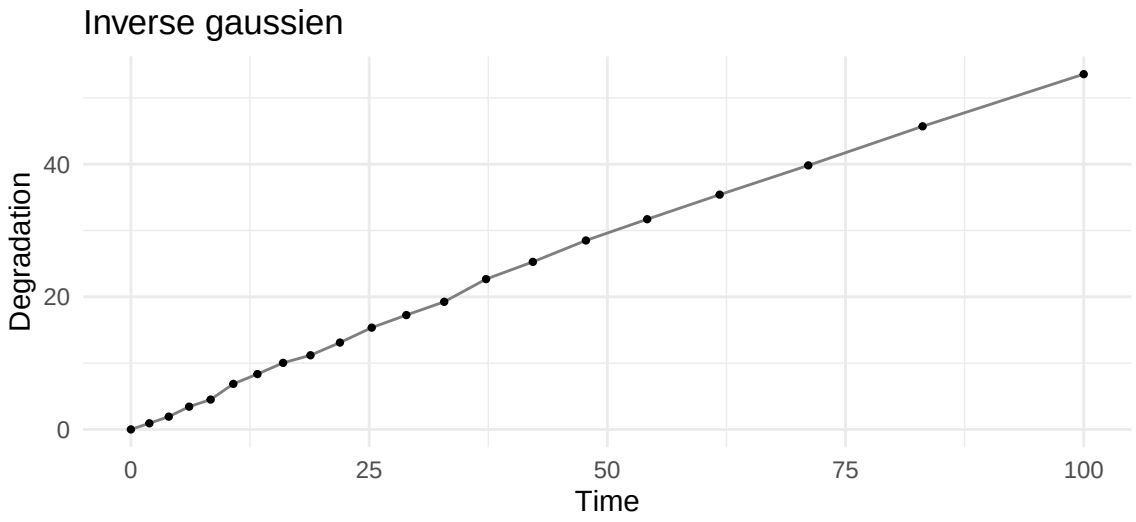
```
set.seed(33)
x <- sim_ig(t3, mu = 0.5, eta = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  1.940035 0.9238276
## [3,]  3.974743 1.9267367
## [4,]  6.113840 3.4350758
## [5,]  8.368618 4.5045984
## [6,] 10.752309 6.8688533
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.924 1.003 1.508 1.070 2.364 1.478 1.693 1.144 1.914 2.252 1.896 2.002
## [13] 3.424 2.595 3.231 3.205 3.704 4.415 5.894 7.865
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

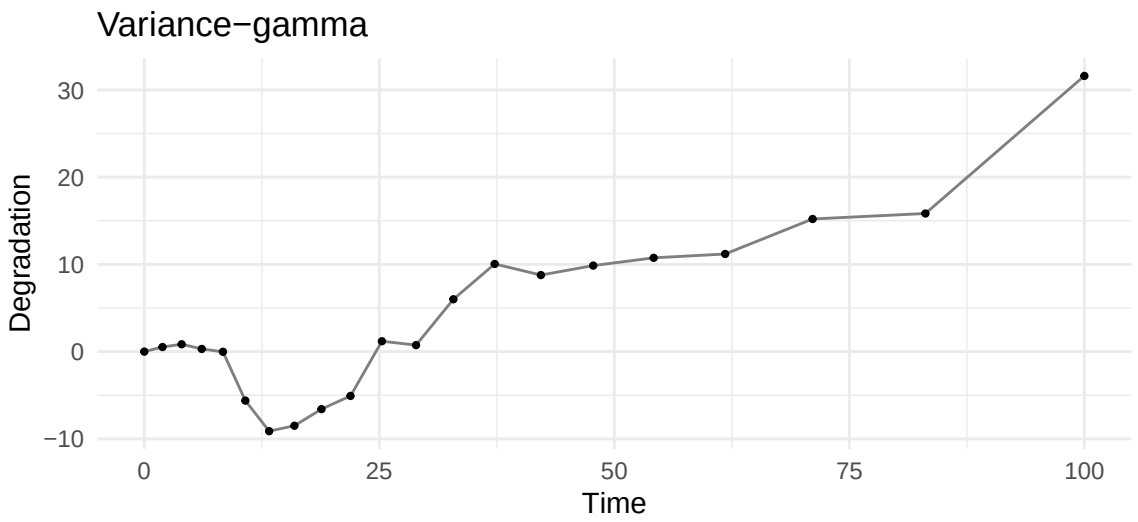
```
set.seed(34)
x <- sim_vg(t3, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t          X1
## [1,]  0.000000  0.00000000
## [2,]  1.940035  0.52692453
## [3,]  3.974743  0.84233589
## [4,]  6.113840  0.30912827
## [5,]  8.368618 -0.01932868
## [6,] 10.752309 -5.61595766
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.527  0.315 -0.533 -0.328 -5.597 -3.498  0.620  1.903  1.513  6.269
## [11] -0.446  5.257  4.048 -1.275  1.086  0.890  0.440  4.011  0.630 15.771
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

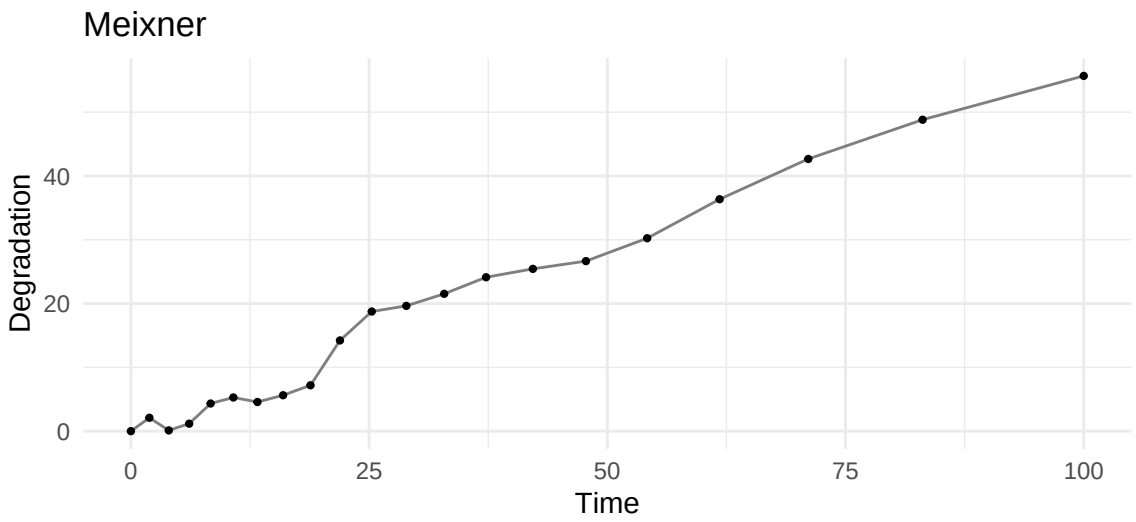
```
set.seed(35)
x <- sim_meixner(t3, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  1.940035 2.0981428
## [3,]  3.974743 0.1163328
## [4,]  6.113840 1.1689796
## [5,]  8.368618 4.3318852
## [6,] 10.752309 5.2801270
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.098 -1.982  1.053  3.163  0.948 -0.713  1.061  1.563  7.025  4.535
## [11]  0.886  1.892  2.595  1.315  1.213  3.589  6.116  6.314  6.149  6.868
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

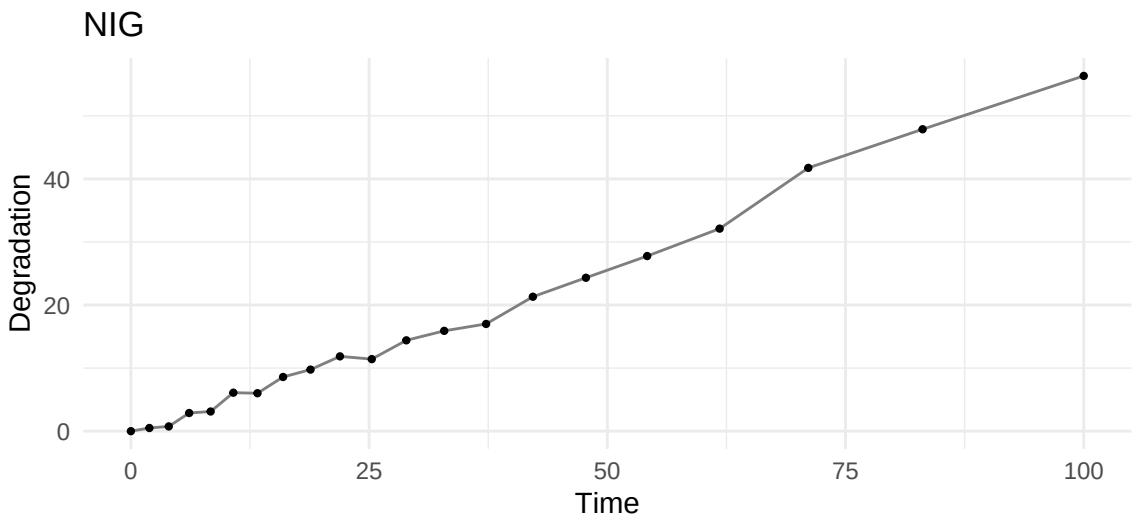
```
set.seed(36)
x <- sim_nig(t3, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  1.940035 0.4992997
## [3,]  3.974743 0.7488852
## [4,]  6.113840 2.8742631
## [5,]  8.368618 3.1134810
## [6,] 10.752309 6.0980221
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.499  0.250  2.125  0.239  2.985 -0.086  2.579  1.169  2.098 -0.438
## [11]  2.981  1.503  1.094  4.313  3.011  3.444  4.345  9.634  6.128  8.460
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

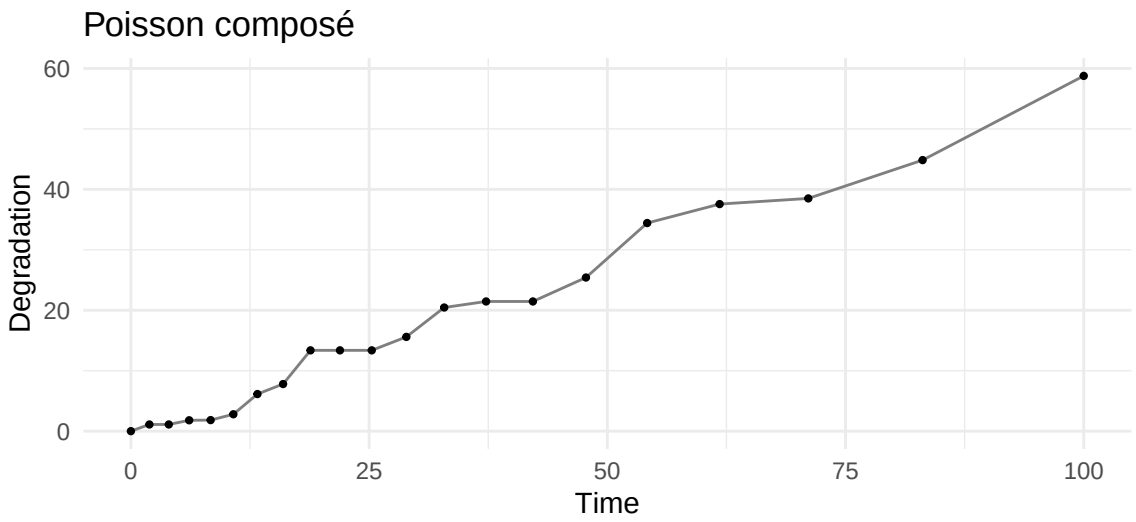
```
set.seed(37)
x <- sim_cpoisson(t3, lambda = 0.5)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  1.940035 1.104451
## [3,]  3.974743 1.104451
## [4,]  6.113840 1.796155
## [5,]  8.368618 1.827021
## [6,] 10.752309 2.792182
```

```
round(diff(x[, "X1"]), 3)
```

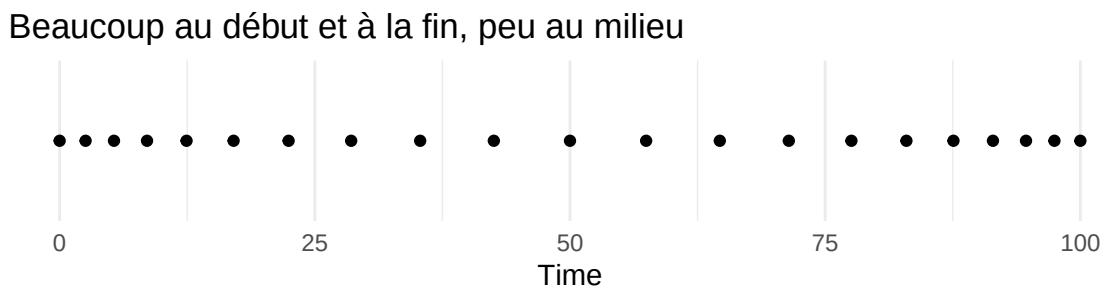
```
##  [1]  1.104  0.000  0.692  0.031  0.965  3.344  1.657  5.574  0.000  0.000
## [11]  2.224  4.862  0.999  0.000  3.960  9.014  3.136  0.937  6.341 13.930
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
Time
0
20
40
60
0
25
50
75
100

Cas 4 : mixtes, beaucoup au début et à la fin

```
t4 <- inspection_times("inc-dec", n = 20, tmax = 100)$time
plot_plan(t4, "Beaucoup au début et à la fin, peu au milieu")
```



Modèle 1 : gamma

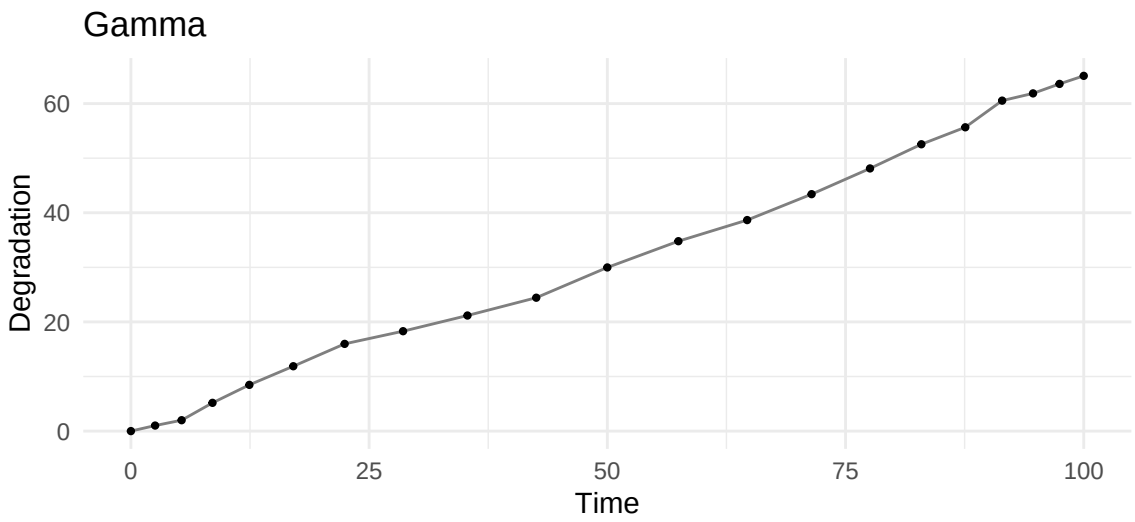
```
set.seed(41)
x <- sim_gamma(t4, a = 2, b = 3)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  2.540921  1.013078
## [3,]  5.322554  1.993097
## [4,]  8.562047  5.174992
## [5,] 12.431733  8.468824
## [6,] 17.042253 11.882103
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.013 0.980 3.182 3.294 3.413 4.097 2.322 2.872 3.263 5.544 4.811 3.869
## [13] 4.746 4.715 4.429 3.110 4.884 1.327 1.738 1.476
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

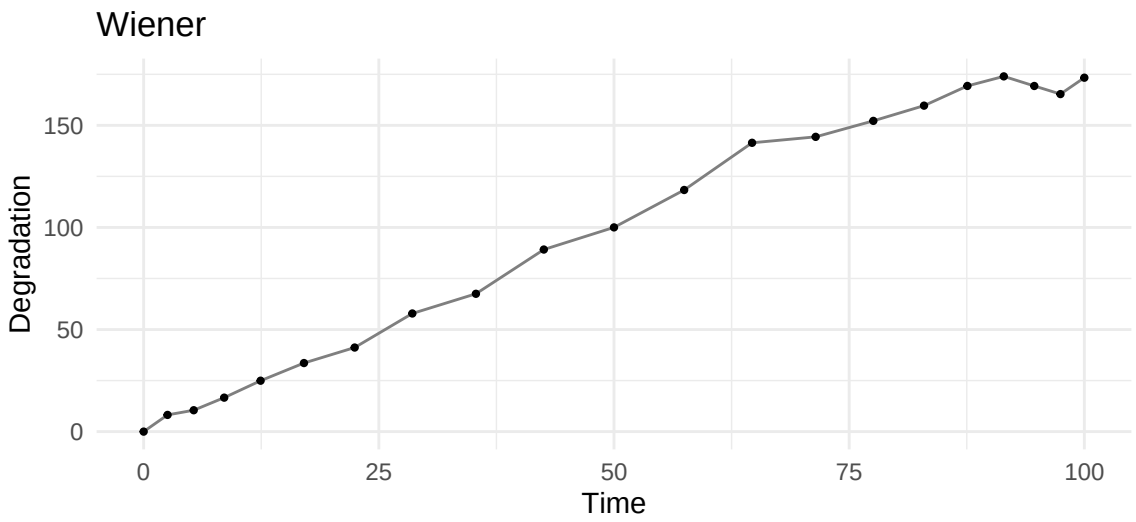
```
set.seed(42)
x <- sim_wiener(t4, mu = 1.5, sigma = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000  0.00000
## [2,]  2.540921  8.18207
## [3,]  5.322554 10.47089
## [4,]  8.562047 16.63729
## [5,] 12.431733 24.93169
## [6,] 17.042253 33.58357
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  8.182  2.289  6.166  8.294  8.652  7.591 16.680  9.649 21.673 10.846
## [11] 18.316 23.115  2.918  7.815  7.465  9.647  4.686 -4.703 -3.968  8.020
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

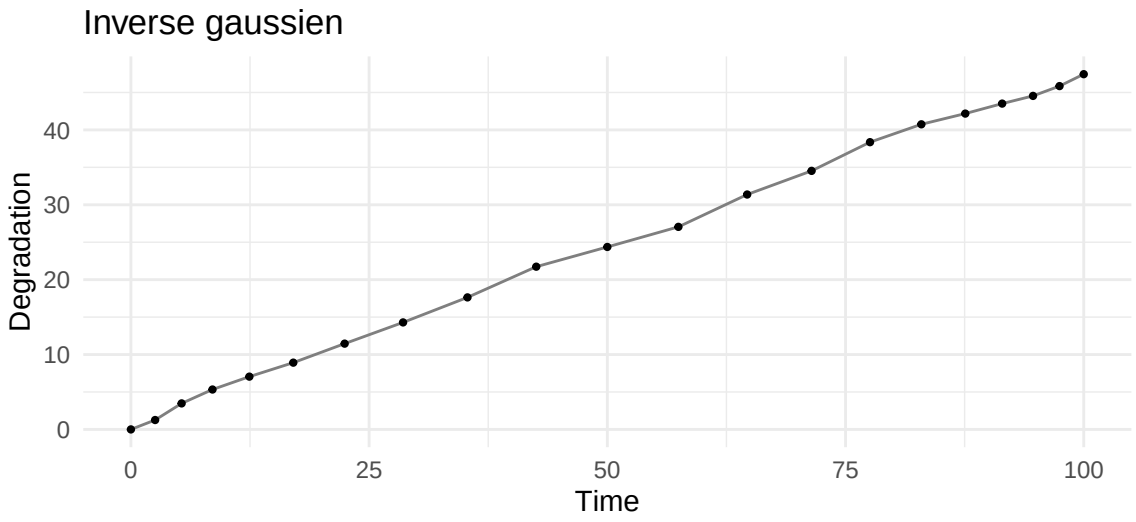
```
set.seed(43)
x <- sim_ig(t4, mu = 0.5, eta = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  2.540921 1.255599
## [3,]  5.322554 3.475956
## [4,]  8.562047 5.329629
## [5,] 12.431733 7.048824
## [6,] 17.042253 8.917167
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.256 2.220 1.854 1.719 1.868 2.538 2.835 3.341 4.100 2.635 2.685 4.320
## [13] 3.158 3.831 2.385 1.432 1.335 1.028 1.309 1.592
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

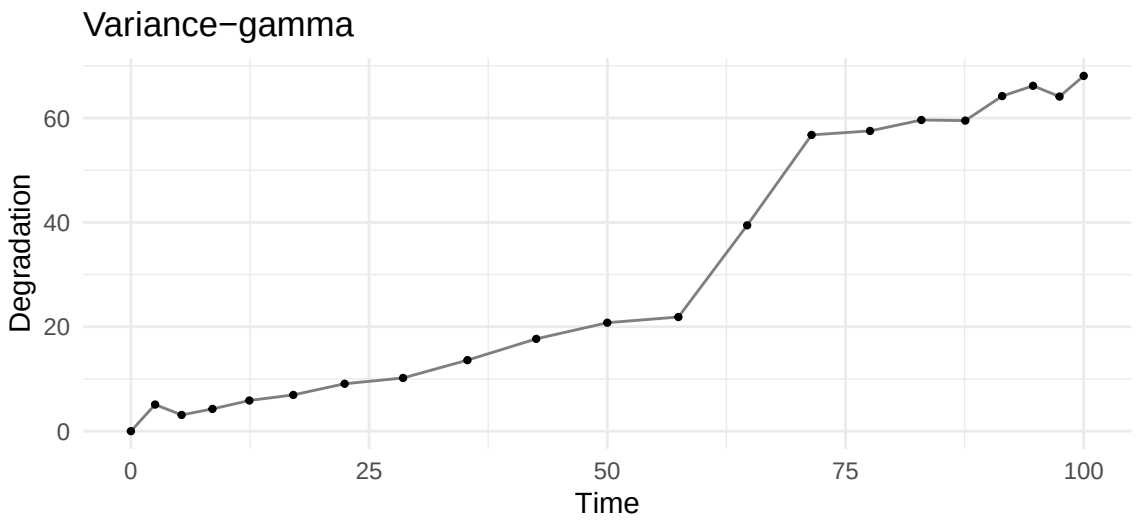
```
set.seed(44)
x <- sim_vg(t4, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  2.540921 5.095509
## [3,]  5.322554 3.104549
## [4,]  8.562047 4.258139
## [5,] 12.431733 5.869560
## [6,] 17.042253 6.941949
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.096 -1.991  1.154  1.611  1.072  2.137  1.112  3.414  4.080  3.092
## [11]  1.093 17.575 17.326  0.764  2.093 -0.115  4.696  1.969 -2.059  3.960
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

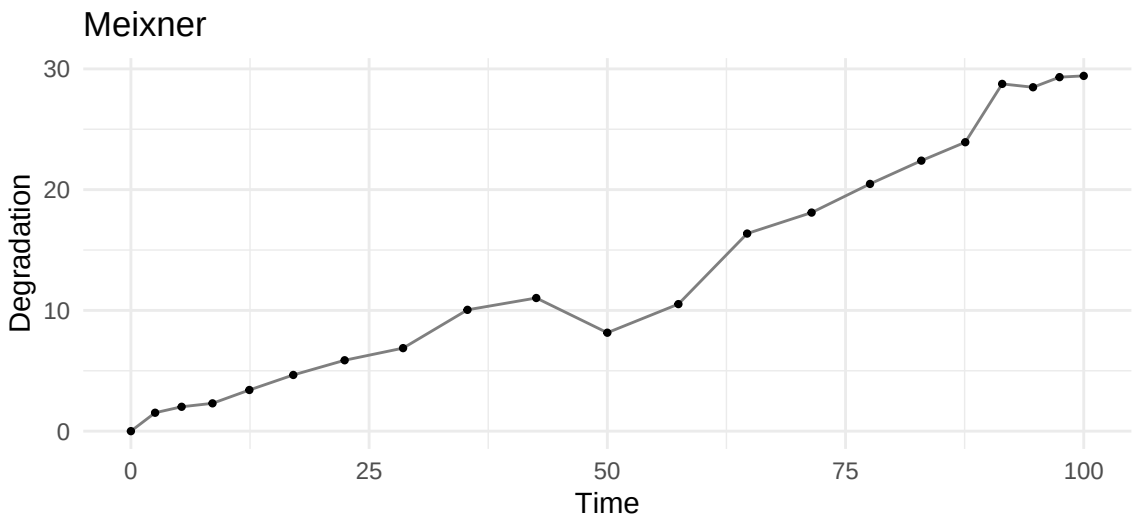
```
set.seed(45)
x <- sim_meixner(t4, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  2.540921 1.523821
## [3,]  5.322554 2.017361
## [4,]  8.562047 2.301981
## [5,] 12.431733 3.408298
## [6,] 17.042253 4.652765
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.524  0.494  0.285  1.106  1.244  1.217  1.005  3.171  0.980 -2.874
## [11]  2.368  5.843  1.737  2.374  1.926  1.526  4.830 -0.278  0.837  0.103
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

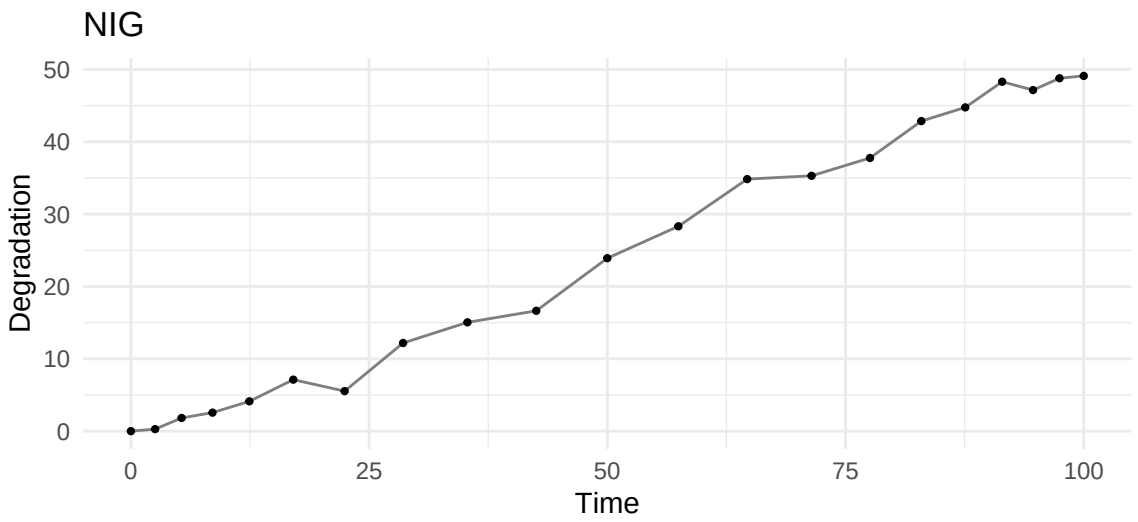
```
set.seed(46)
x <- sim_nig(t4, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  2.540921 0.2773168
## [3,]  5.322554 1.8228331
## [4,]  8.562047 2.5582928
## [5,] 12.431733 4.1226725
## [6,] 17.042253 7.1190832
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.277  1.546  0.735  1.564  2.996 -1.589  6.654  2.858  1.593  7.274
## [11]  4.408  6.528  0.456  2.463  5.101  1.895  3.545 -1.151  1.634  0.325
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

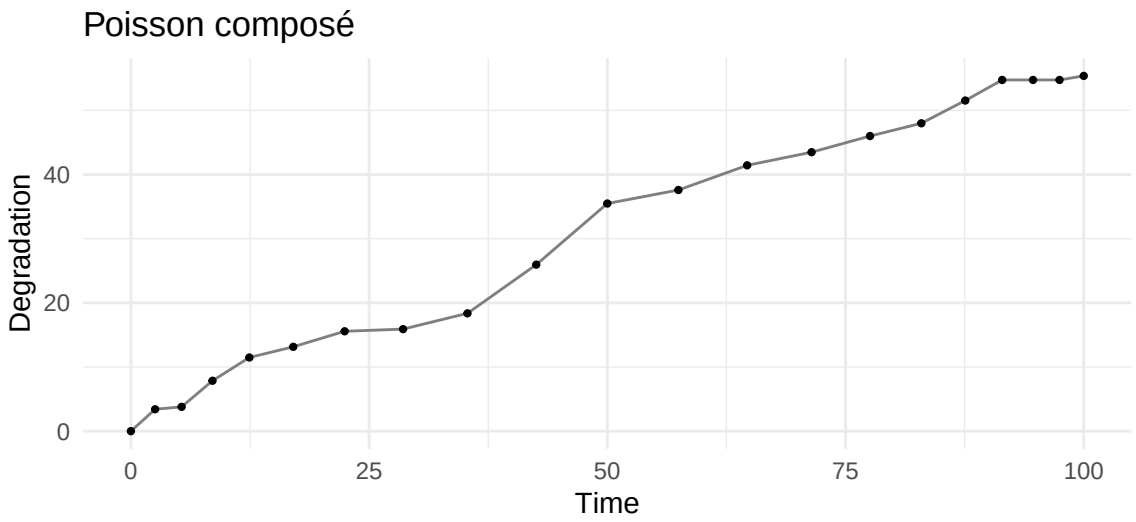
```
set.seed(47)
x <- sim_cpoisson(t4, lambda = 0.5)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  2.540921  3.410377
## [3,]  5.322554  3.788944
## [4,]  8.562047  7.855023
## [5,] 12.431733 11.480214
## [6,] 17.042253 13.146896
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.410 0.379 4.066 3.625 1.667 2.421 0.337 2.461 7.588 9.529 2.107 3.847
## [13] 2.046 2.519 1.991 3.545 3.210 0.000 0.000 0.631
```

```
plot_traj(x, "Poisson composé")
```


Poisson composé
Degradation
0
20
40
0
25
50
75
100
Time

Cas 5 : mixtes, beaucoup au milieu

```
t5 <- inspection_times("dec-inc", n = 20, tmax = 100)$time
plot_plan(t5, "Peu au début et à la fin, beaucoup au milieu")
```



Modèle 1 : gamma

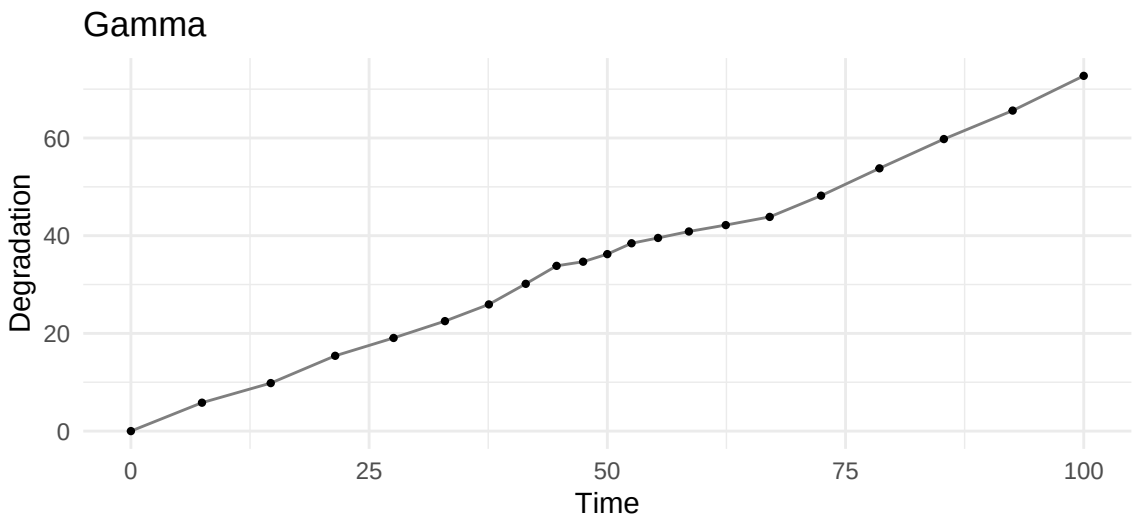
```
set.seed(51)
x <- sim_gamma(t5, a = 2, b = 3)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  5.814275
## [3,] 14.677446  9.824488
## [4,] 21.437953 15.412420
## [5,] 27.568267 19.065485
## [6,] 32.957747 22.511943
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 5.814 4.010 5.588 3.653 3.446 3.420 4.216 3.677 0.845 1.554 2.221 1.091
## [13] 1.322 1.314 1.673 4.344 5.600 5.990 5.831 7.110
```

```
plot_traj(x, "Gamma")
```



Modèle 2 : Wiener

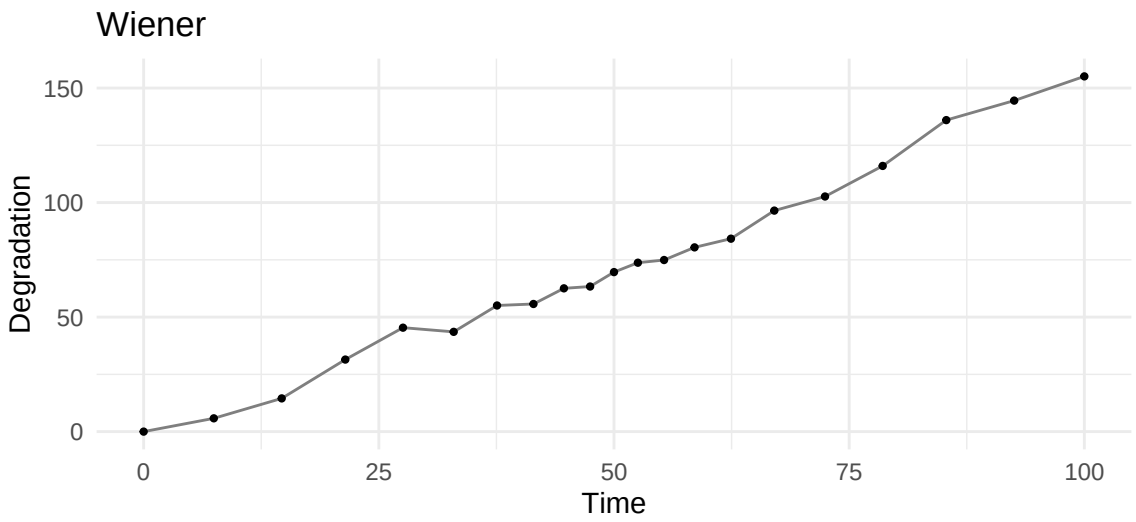
```
set.seed(52)
x <- sim_wiener(t5, mu = 1.5, sigma = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  5.801785
## [3,] 14.677446 14.507619
## [4,] 21.437953 31.452913
## [5,] 27.568267 45.376256
## [6,] 32.957747 43.556482
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.802  8.706 16.945 13.923 -1.820 11.506  0.661  6.829  0.794  6.298
## [11]  4.099  1.176  5.517  3.794 12.261  6.155 13.353 20.002  8.512 10.582
```

```
plot_traj(x, "Wiener")
```



Modèle 3 : inverse gaussien

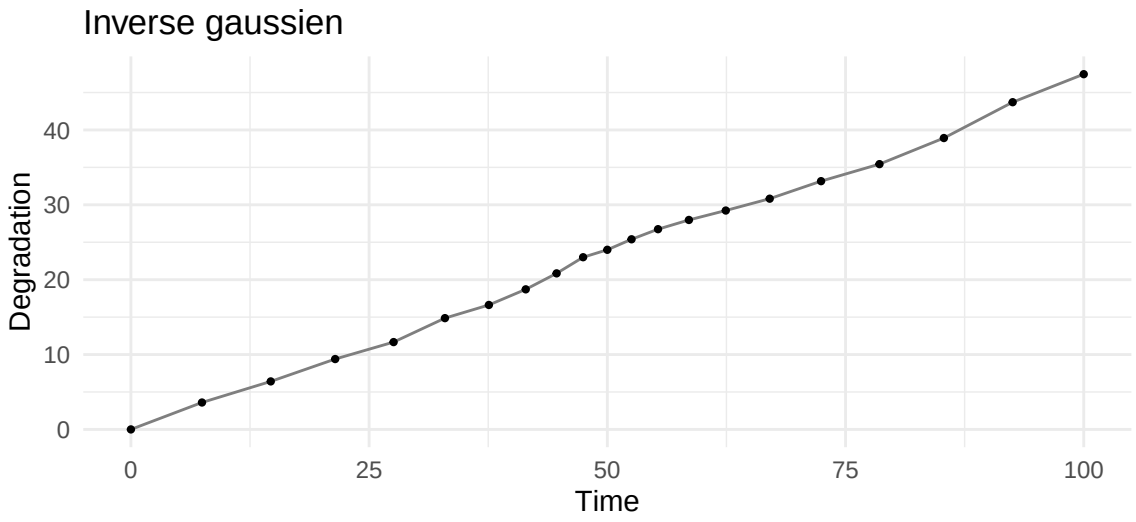
```
set.seed(53)
x <- sim_ig(t5, mu = 0.5, eta = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  3.595579
## [3,] 14.677446  6.412297
## [4,] 21.437953  9.399752
## [5,] 27.568267 11.663964
## [6,] 32.957747 14.863754
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.596 2.817 2.987 2.264 3.200 1.759 2.088 2.132 2.158 0.992 1.403 1.350
## [13] 1.232 1.260 1.581 2.342 2.275 3.485 4.787 3.742
```

```
plot_traj(x, "Inverse gaussien")
```



Modèle 4 : variance-gamma

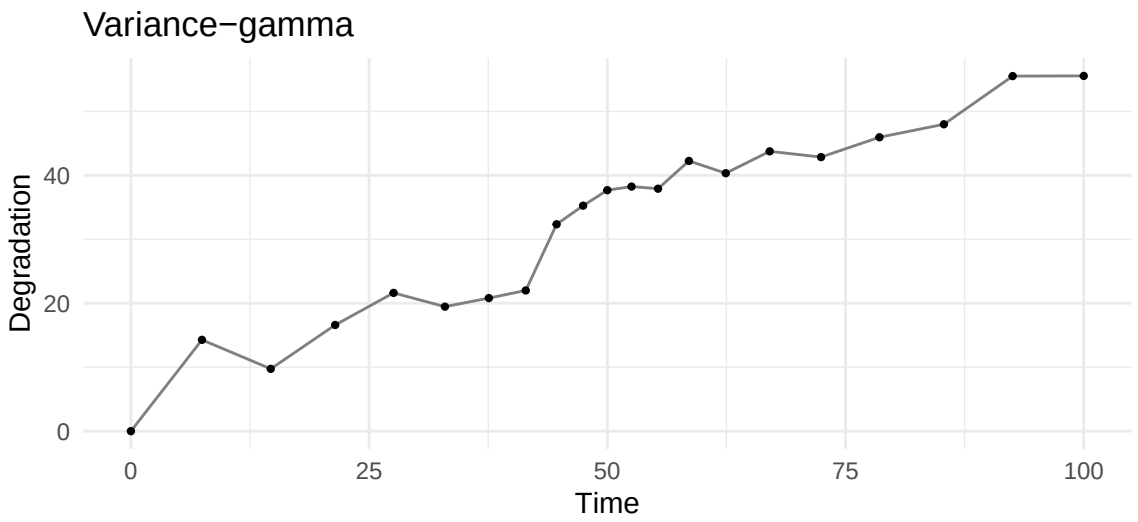
```
set.seed(54)
x <- sim_vg(t5, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079 14.277776
## [3,] 14.677446  9.742506
## [4,] 21.437953 16.599526
## [5,] 27.568267 21.619767
## [6,] 32.957747 19.463350
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 14.278 -4.535  6.857  5.020 -2.156  1.346  1.193 10.348  2.910  2.420
## [11]  0.563 -0.331  4.342 -1.921  3.421 -0.900  3.103  2.027  7.536  0.031
```

```
plot_traj(x, "Variance-gamma")
```



Modèle 5 : Meixner

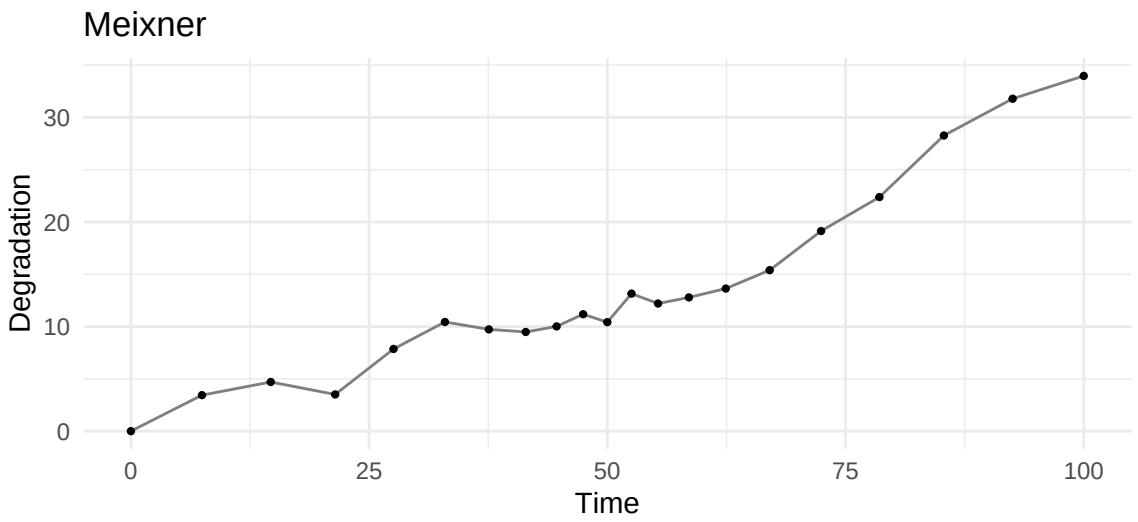
```
set.seed(55)
x <- sim_meixner(t5, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  3.442434
## [3,] 14.677446  4.697757
## [4,] 21.437953  3.507490
## [5,] 27.568267  7.857074
## [6,] 32.957747 10.436038
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.442  1.255 -1.190  4.350  2.579 -0.708 -0.247  0.531  1.165 -0.758
## [11]  2.727 -0.946  0.586  0.846  1.759  3.740  3.247  5.884  3.518  2.182
```

```
plot_traj(x, "Meixner")
```



Modèle 6 : normal inverse gaussien (NIG)

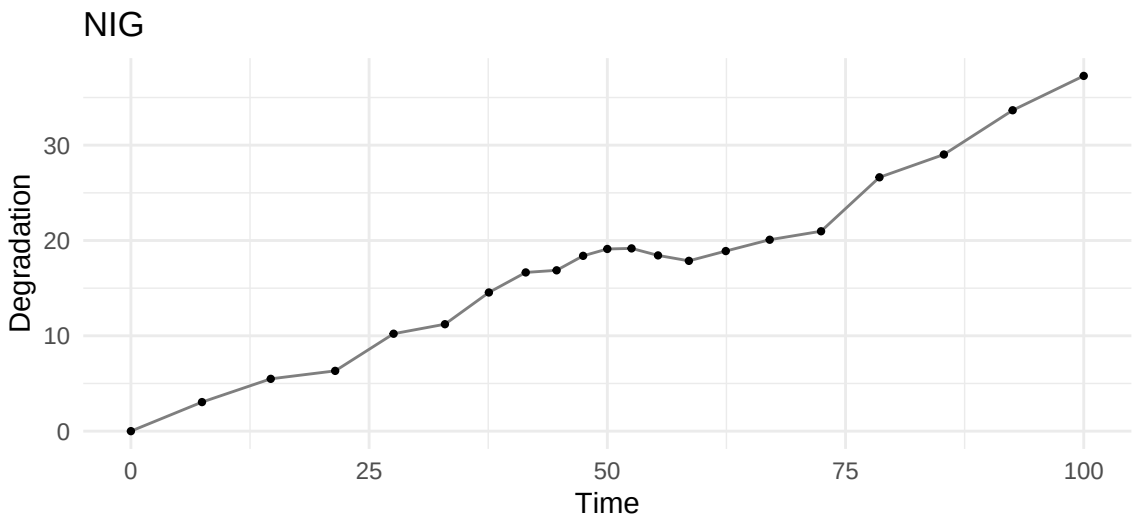
```
set.seed(56)
x <- sim_nig(t5, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.459079  3.047164
## [3,] 14.677446  5.487323
## [4,] 21.437953  6.322425
## [5,] 27.568267 10.219941
## [6,] 32.957747 11.218525
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.047  2.440  0.835  3.898  0.999  3.328  2.105  0.220  1.519  0.720
## [11]  0.061 -0.735 -0.571  1.027  1.187  0.890  5.661  2.391  4.643  3.604
```

```
plot_traj(x, "NIG")
```



Modèle 7 : Poisson composé

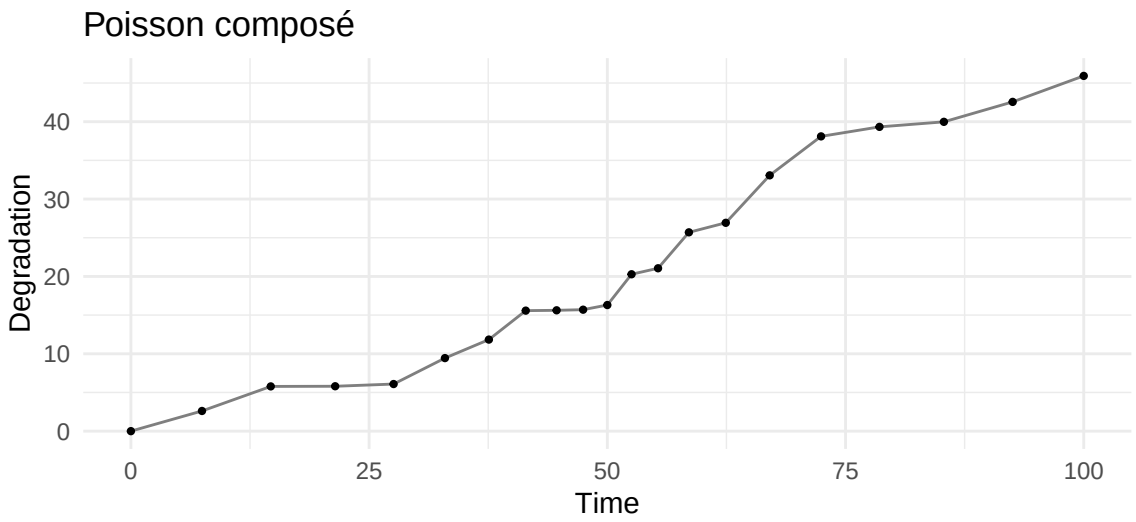
```
set.seed(57)
x <- sim_cpoisson(t5, lambda = 0.5)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  7.459079 2.606789
## [3,] 14.677446 5.781682
## [4,] 21.437953 5.802607
## [5,] 27.568267 6.088267
## [6,] 32.957747 9.442441
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.607 3.175 0.021 0.286 3.354 2.392 3.743 0.043 0.080 0.602 3.977 0.774
## [13] 4.646 1.231 6.136 5.036 1.230 0.651 2.577 3.361
```

```
plot_traj(x, "Poisson composé")
```

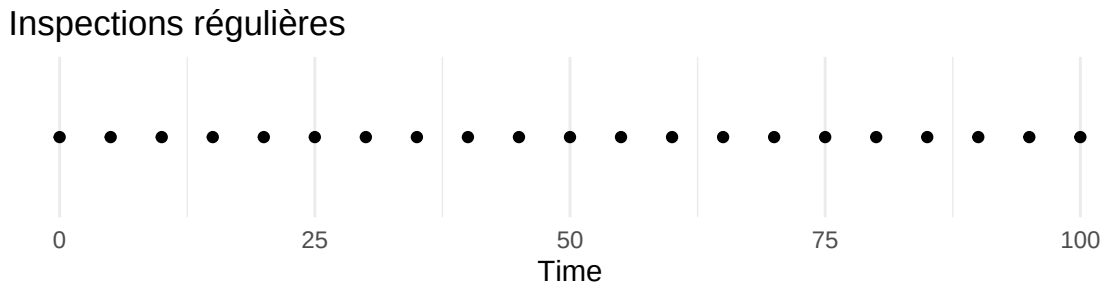

Poisson composé
Degradation
0
10
20
30
40
0
25
50
75
100
Time

II - Plusieurs trajectoires

5 trajectoires, chacune inspectée 20 fois entre 0 et 100, aux mêmes instants. `x` est une matrice : une colonne `t`, puis `X1`, ..., `X5`. On affiche les premières lignes des trois premières trajectoires.

Cas 1 : inspections régulières

```
p1 <- inspection_times("regular", n = 20, n_traj = 5, tmax = 100)
plot_plan(p1$time[p1$id == 1], "Inspections régulières")
```



Modèle 1 : gamma

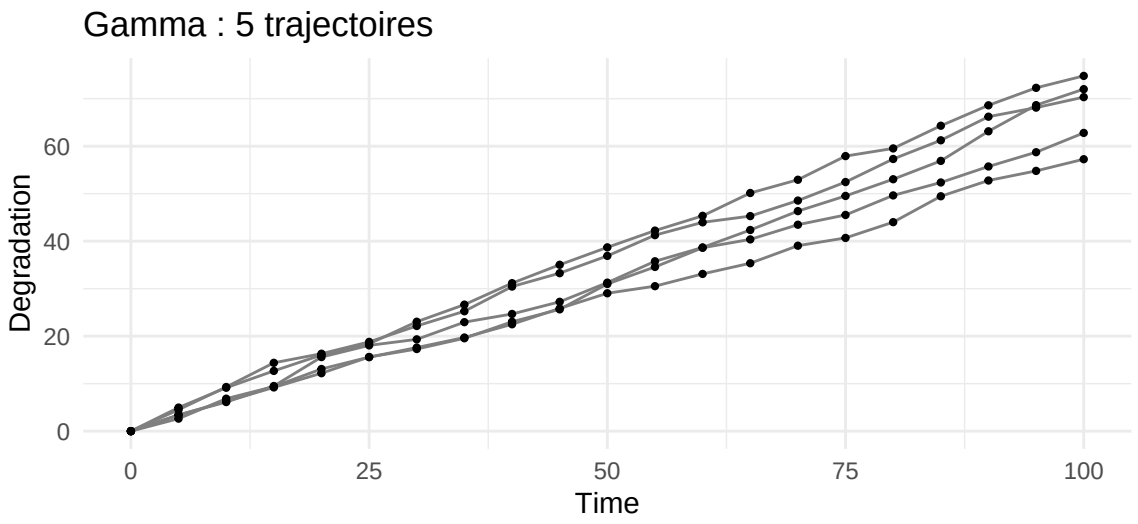
```
set.seed(111)
x <- sim_gamma(p1, a = 2, b = 3)
head(x[, 1:4])
```

```
##       t        X1        X2        X3
## [1,]  0  0.000000  0.000000  0.000000
## [2,]  5  3.412944  3.391215  4.962620
## [3,] 10  6.248927  6.129922  9.138235
## [4,] 15  9.207337  9.496121 12.684072
## [5,] 20 12.200879 15.598724 16.033187
## [6,] 25 15.606267 18.057525 18.389191
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.413 2.836 2.958 2.994 3.405 1.684 2.267 3.473 2.611 5.347 3.584 4.039
## [13] 1.762 3.088 2.059 4.112 2.727 3.365 3.005 4.065
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

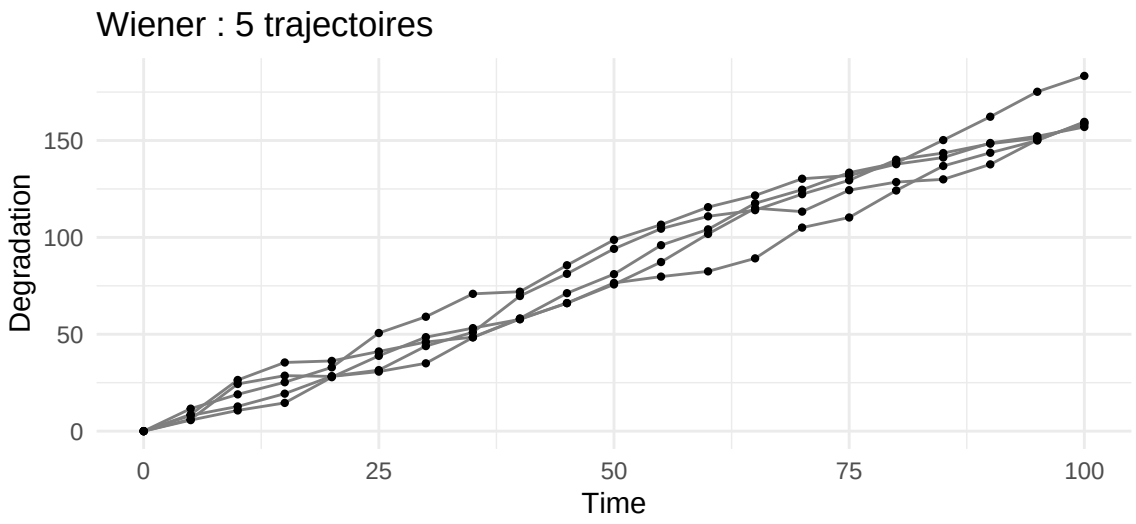
```
set.seed(112)
x <- sim_wiener(p1, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##       t        X1        X2       X3
## [1,]  0  0.000000  0.000000  0.00000
## [2,]  5  6.094713  8.613025 11.60806
## [3,] 10 24.342933 26.400095 18.97625
## [4,] 15 28.630759 35.438839 25.23648
## [5,] 20 28.257068 36.240624 32.90835
## [6,] 25 30.724657 41.112867 50.63640
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  6.095 18.248  4.288 -0.374  2.468  4.282 13.359  9.521  8.181 10.422
## [11]  3.263  2.710  6.713 15.872  5.218 13.935 12.657  6.808  6.338  9.608
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

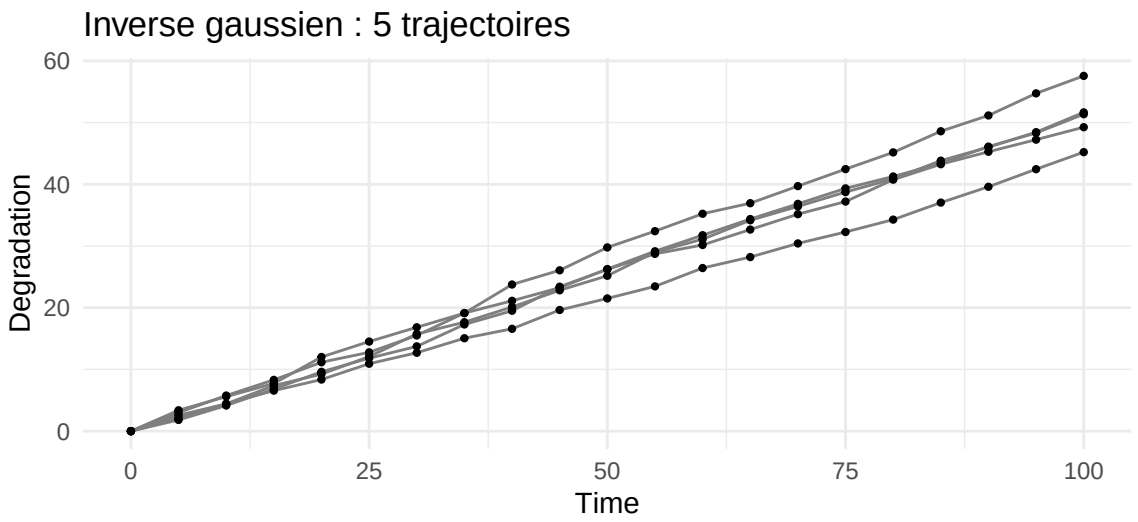
```
set.seed(113)
x <- sim_ig(p1, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##       t        X1        X2        X3
## [1,]  0  0.000000  0.000000  0.000000
## [2,]  5  2.575667  1.816995  3.380367
## [3,] 10  4.416063  4.148258  5.600797
## [4,] 15  7.371107  6.834152  7.721204
## [5,] 20  9.244923  9.608309 12.018158
## [6,] 25 12.077408 11.820806 14.513946
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.576 1.840 2.955 1.874 2.832 3.674 1.901 2.465 2.708 2.349 3.822 2.106
## [13] 3.062 2.242 2.329 2.141 2.951 2.186 2.432 3.209
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

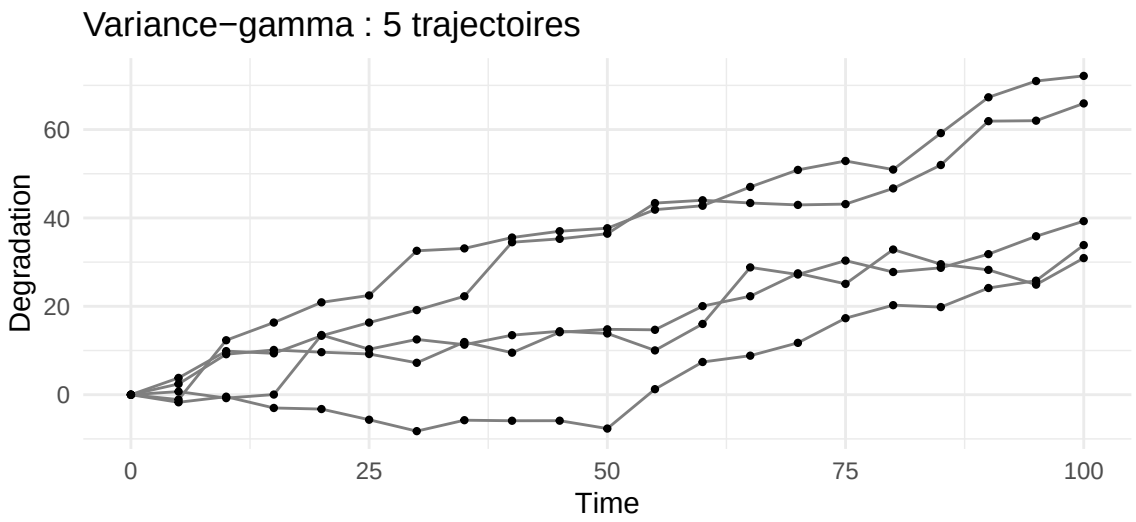
```
set.seed(114)
x <- sim_vg(p1, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##       t        X1          X2        X3
## [1,]  0  0.000000  0.00000000  0.000000
## [2,]  5  2.436627  0.70296417 -1.107265
## [3,] 10  9.162189 -0.77011502 12.323899
## [4,] 15 10.122663  0.03664116 16.329080
## [5,] 20  9.624831 13.50917123 20.901106
## [6,] 25  9.211656 10.28751843 22.462301
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.437  6.726  0.960 -0.498 -0.413 -1.990  4.699 -2.413  4.626  0.661
## [11] -0.109  5.361  2.242  5.178 -2.377  7.766 -3.320 -1.284 -3.340  6.006
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

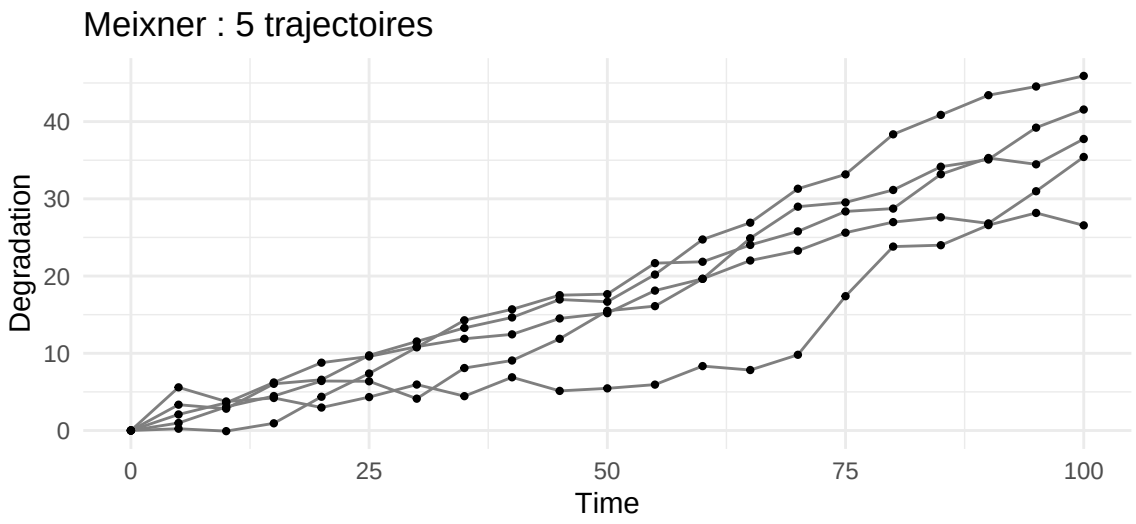
```
set.seed(115)
x <- sim_meixner(p1, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##       t       X1       X2          X3
## [1,]  0 0.000000 0.000000  0.00000000
## [2,]  5 3.339643 2.078708  0.22800433
## [3,] 10 2.823348 3.582343 -0.08350747
## [4,] 15 6.044890 6.217424  0.93099359
## [5,] 20 6.571270 8.782078  4.36570414
## [6,] 25 9.730140 9.586900  7.38028101
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.340 -0.516  3.222  0.526  3.159  1.800  1.755  1.354  2.323 -0.290
## [11]  3.520  4.541  2.174  4.399  1.858  5.186  2.518  2.551  1.127  1.372
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

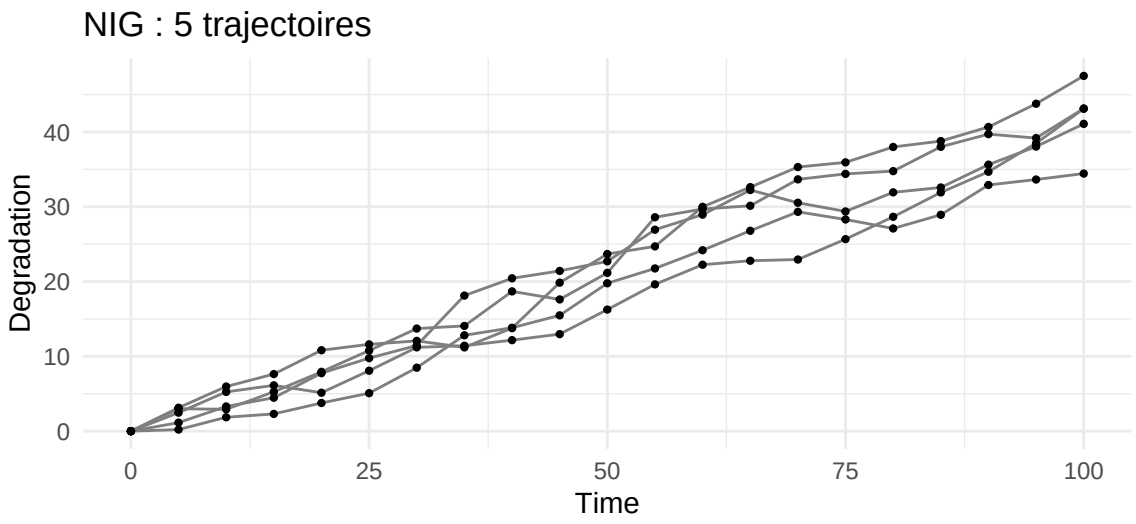
```
set.seed(116)
x <- sim_nig(p1, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##       t       X1        X2        X3
## [1,]  0 0.000000  0.000000 0.0000000
## [2,]  5 2.468533  2.990295 0.2074206
## [3,] 10 5.246575  2.923195 1.8555426
## [4,] 15 6.122326  5.263634 2.3019116
## [5,] 20 5.136418  7.929152 3.7644948
## [6,] 25 8.077892 10.777178 5.0749732
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.469  2.778  0.876 -0.986  2.941  3.123  0.221  0.743  0.804  3.282
## [11]  3.372  2.628  0.529  0.166  2.722  2.988  3.257  2.767  3.812  4.617
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

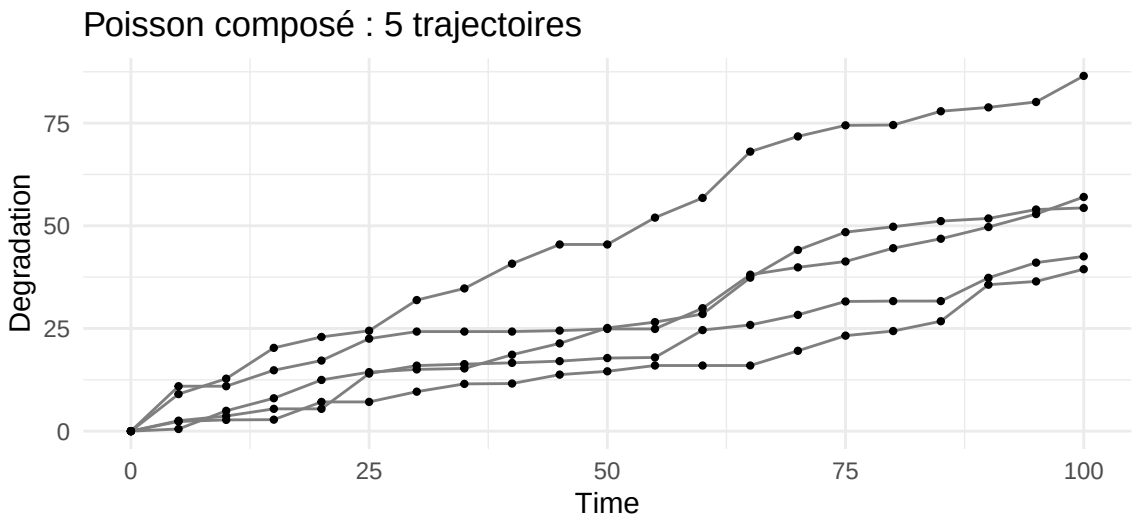
```
set.seed(117)
x <- sim_cpoisson(p1, lambda = 0.5)
head(x[, 1:4])
```

```
##       t       X1       X2        X3
## [1,]  0  0.00000  0.00000  0.000000
## [2,]  5 10.93951  9.02175  2.547582
## [3,] 10 10.93951 12.77248  3.677322
## [4,] 15 14.81978 20.27313  5.428941
## [5,] 20 17.18333 22.93611  5.428941
## [6,] 25 22.51752 24.44392 14.017366
```

```
round(diff(x[, "X1"]), 3)
```

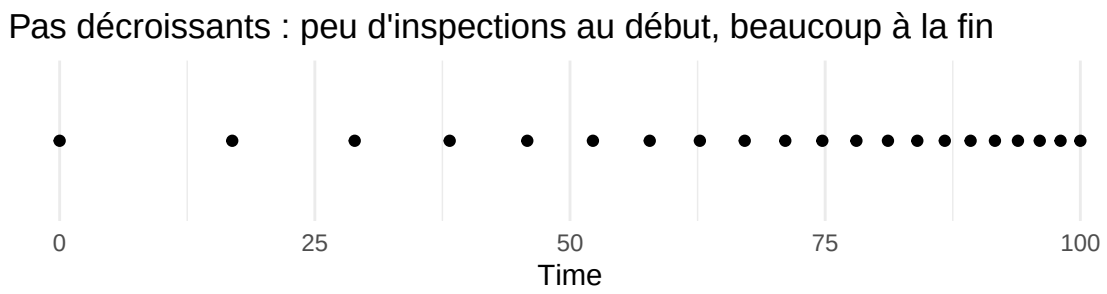
```
##  [1] 10.940  0.000  3.880  2.364  5.334  1.725  0.000  0.000  0.238  0.407
## [11]  0.000  5.049  8.156  1.785  1.432  3.239  2.295  2.852  3.165  4.153
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```


Poisson composé : 5 trajectoires
Degradation
0
25
50
75
0
25
50
75
100
Time

Cas 2 : pas décroissants

```
p2 <- inspection_times("decreasing", n = 20, n_traj = 5, tmax = 100)
plot_plan(p2$time[p2$id == 1], "Pas décroissants : peu d'inspections au début, beaucoup à la fin")
```



Modèle 1 : gamma

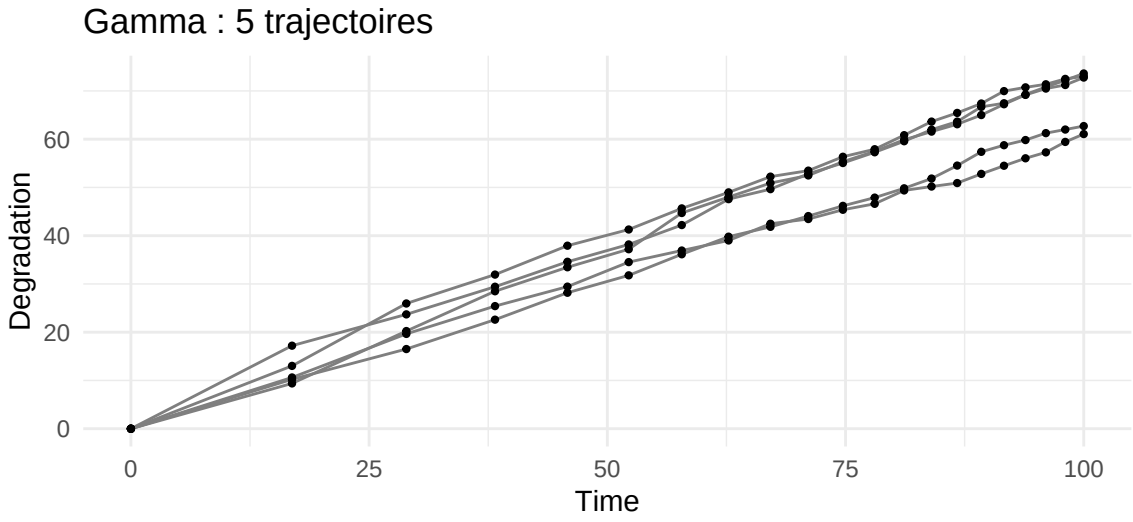
```
set.seed(121)
x <- sim_gamma(p2, a = 2, b = 3)
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.000000
## [2,] 16.90921 10.62024 17.20485  9.386244
## [3,] 28.90648 19.65015 23.69326 20.236937
## [4,] 38.21229 25.40798 29.41898 28.486875
## [5,] 45.81569 29.44038 34.59453 33.456933
## [6,] 52.24427 34.52621 38.18268 37.192447
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 10.620  9.030  5.758  4.032  5.086  2.398  2.089  3.466  0.972  1.918
## [11]  1.235  2.772  0.809  0.720  1.897  1.689  1.545  1.233  2.161  1.626
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

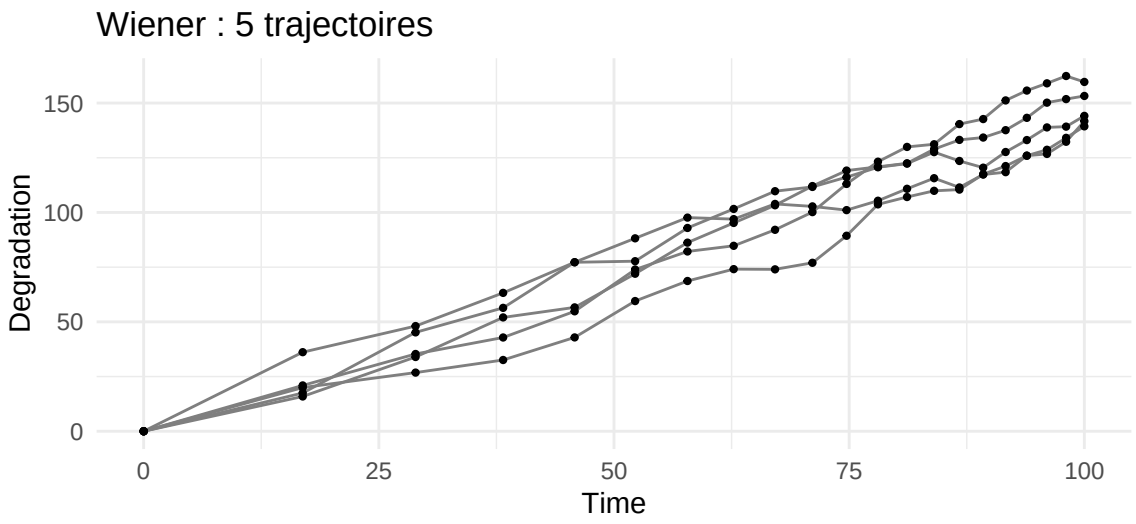
```
set.seed(122)
x <- sim_wiener(p2, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.00000
## [2,] 16.90921 36.14323 20.94065 19.95223
## [3,] 28.90648 48.07174 35.31596 26.80538
## [4,] 38.21229 63.24777 42.85070 32.56368
## [5,] 45.81569 77.22253 54.81566 42.88415
## [6,] 52.24427 77.72730 73.90821 59.50291
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 36.143 11.929 15.176 13.975  0.505 15.191  8.693  8.108  1.947  4.464
## [11]  4.524  1.677  5.246 -4.043 -3.056  7.155  5.454  5.783  0.361  4.913
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

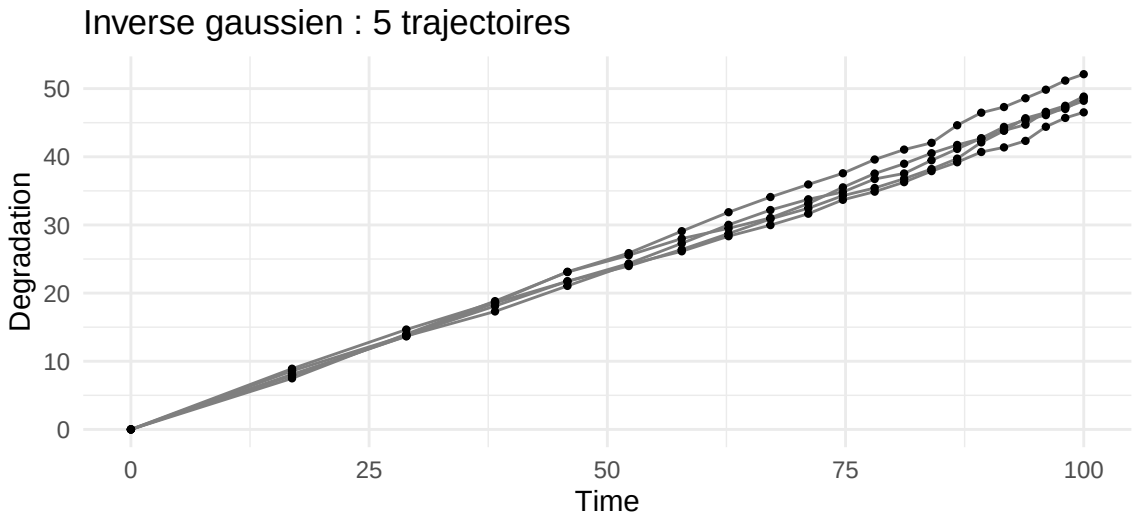
```
set.seed(123)
x <- sim_ig(p2, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##             t        X1        X2        X3
## [1,]  0.00000  0.000000  0.000000  0.000000
## [2,] 16.90921  7.897723  8.904532  8.073231
## [3,] 28.90648 13.700327 14.653043 13.652377
## [4,] 38.21229 17.306694 18.671536 18.566432
## [5,] 45.81569 21.060098 23.128694 21.727826
## [6,] 52.24427 24.193476 25.862694 24.330757
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 7.898 5.803 3.606 3.753 3.133 1.940 2.214 1.627 1.673 2.039 1.196 1.395
## [13] 1.619 1.301 1.505 0.673 0.955 2.070 1.300 0.819
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

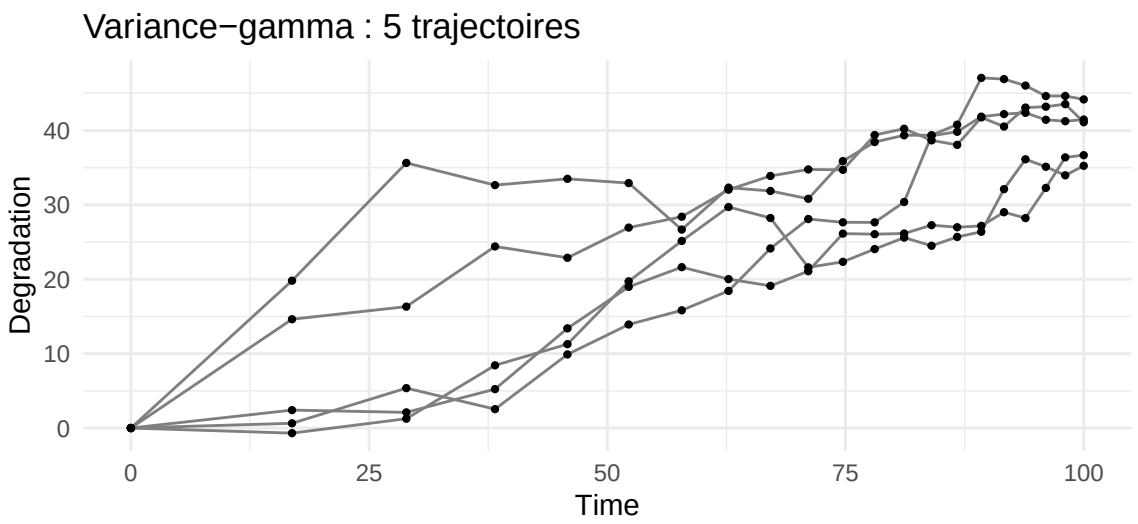
```
set.seed(124)
x <- sim_vg(p2, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##             t        X1        X2       X3
## [1,]  0.00000  0.000000  0.000000  0.00000
## [2,] 16.90921  0.637251  2.409797 19.80395
## [3,] 28.90648  5.363706  2.105576 35.63543
## [4,] 38.21229  2.540566  5.230256 32.65205
## [5,] 45.81569  9.891368 13.399796 33.49435
## [6,] 52.24427 13.917200 18.976941 32.92566
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.637  4.726 -2.823  7.351  4.026  1.908  2.606  5.707  3.958 -0.445
## [11] -0.007  2.737  8.907  0.533  2.031  0.344  0.170 -0.934 -0.195  0.226
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

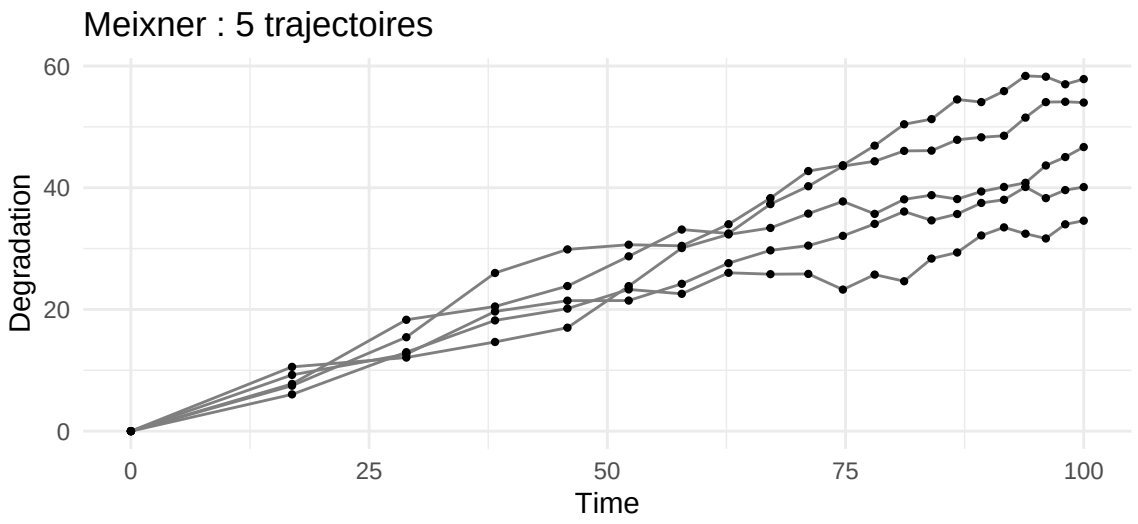
```
set.seed(125)
x <- sim_meixner(p2, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##             t       X1        X2       X3
## [1,]  0.00000  0.00000  0.000000  0.00000
## [2,] 16.90921 10.56185  6.039943  9.24308
## [3,] 28.90648 12.09210 12.976338 12.64901
## [4,] 38.21229 14.65336 18.195877 19.65971
## [5,] 45.81569 16.99743 20.136227 21.44034
## [6,] 52.24427 23.83157 23.285497 21.45917
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 10.562  1.530  2.561  2.344  6.834  6.259  2.230  1.072  2.353  2.001
## [11] -2.044  2.387  0.686 -0.641  1.247  0.744  0.680  2.850  1.385  1.643
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

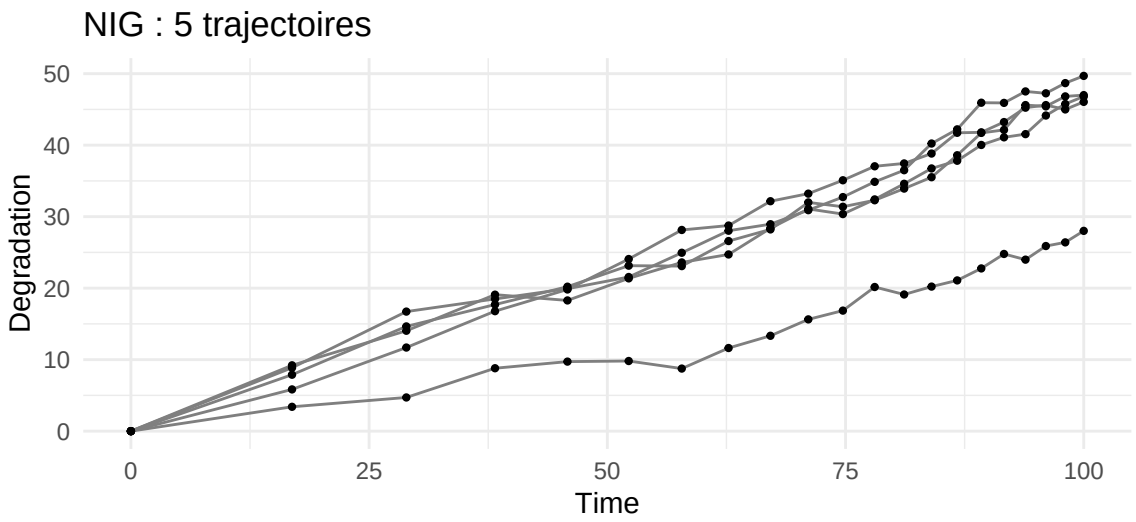
```
set.seed(126)
x <- sim_nig(p2, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##             t       X1        X2        X3
## [1,]  0.00000 0.000000  0.000000  0.000000
## [2,] 16.90921 3.403669  9.210465  8.863915
## [3,] 28.90648 4.711353 14.040403 16.720638
## [4,] 38.21229 8.797905 19.090496 18.494623
## [5,] 45.81569 9.725299 18.282901 19.921991
## [6,] 52.24427 9.809557 21.367375 21.559535
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  3.404  1.308  4.087  0.927  0.084 -1.058  2.875  1.712  2.294  1.226
## [11]  3.301 -1.046  1.104  0.866  1.675  2.024 -0.795  1.903  0.514  1.608
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

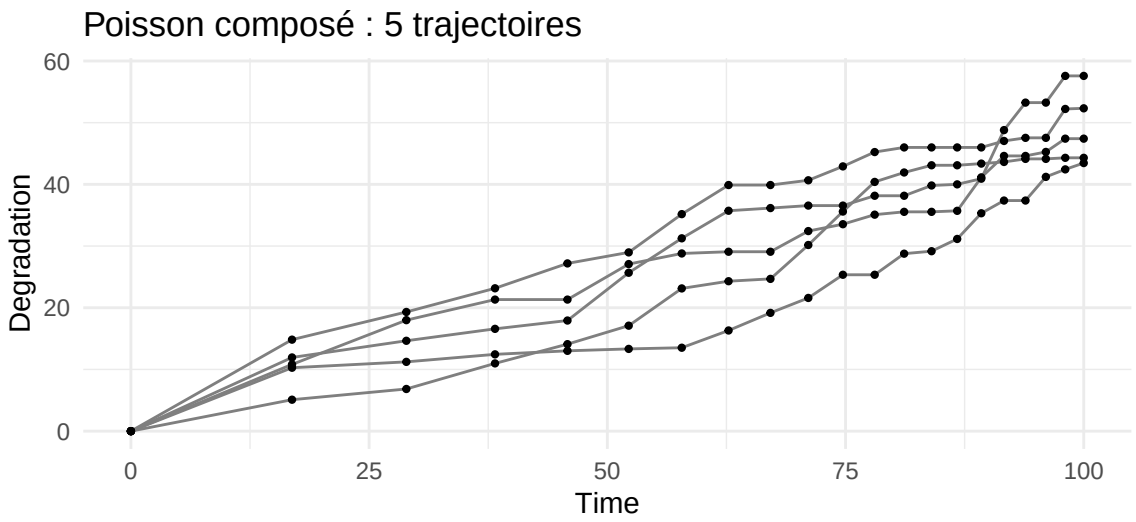
```
set.seed(127)
x <- sim_cpoisson(p2, lambda = 0.5)
head(x[, 1:4])
```

```
##             t        X1       X2       X3
## [1,]  0.00000  0.000000  0.00000  0.00000
## [2,] 16.90921  5.094637 10.27714 14.82955
## [3,] 28.90648  6.832984 11.22789 19.31514
## [4,] 38.21229 10.979469 12.44772 23.15368
## [5,] 45.81569 14.098460 13.01757 27.18597
## [6,] 52.24427 17.094252 13.32985 28.97341
```

```
round(diff(x[, "X1"]), 3)
```

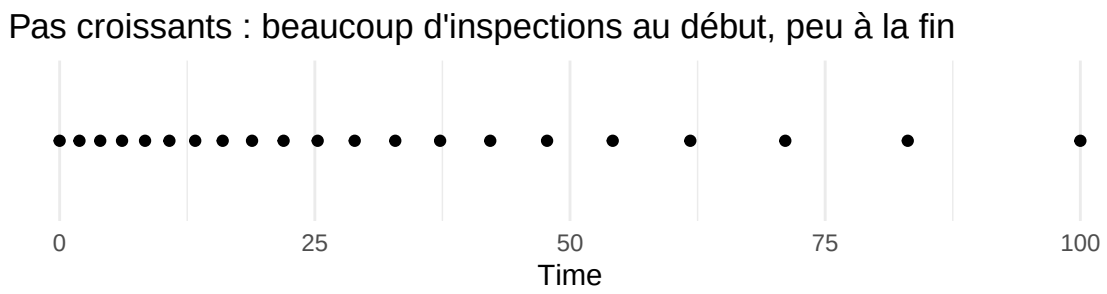
```
##  [1] 5.095 1.738 4.146 3.119 2.996 6.038 1.170 0.377 5.492 5.420 4.804 1.528
## [13] 1.169 0.000 0.274 0.266 0.499 0.000 0.170 0.000
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```


Poisson composé : 5 trajectoires
Degradation
Time
0
20
40
60
0
25
50
75
100

Cas 3 : pas croissants

```
p3 <- inspection_times("increasing", n = 20, n_traj = 5, tmax = 100)
plot_plan(p3$time[p3$id == 1], "Pas croissants : beaucoup d'inspections au début, peu à la fin")
```



Modèle 1 : gamma

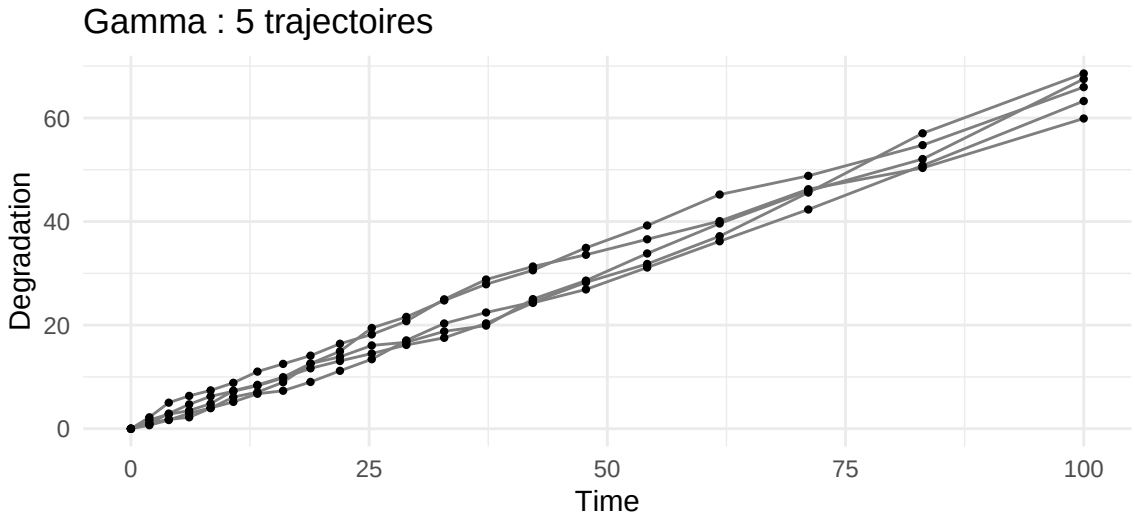
```
set.seed(131)
x <- sim_gamma(p3, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1       X2       X3
## [1,]  0.000000 0.0000000 0.000000 0.000000
## [2,]  1.940035 0.6807504 1.060241 2.206014
## [3,]  3.974743 1.6795349 1.791835 5.027377
## [4,]  6.113840 2.9693747 2.202258 6.336759
## [5,]  8.368618 4.0223494 3.975035 7.405013
## [6,] 10.752309 5.1828148 6.103820 8.862926
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.681  0.999  1.290  1.053  1.160  1.560  0.585  1.699  2.156  2.235
## [11]  3.633  3.264  2.122  1.999  3.815  3.560  5.353  8.435 11.436 11.550
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

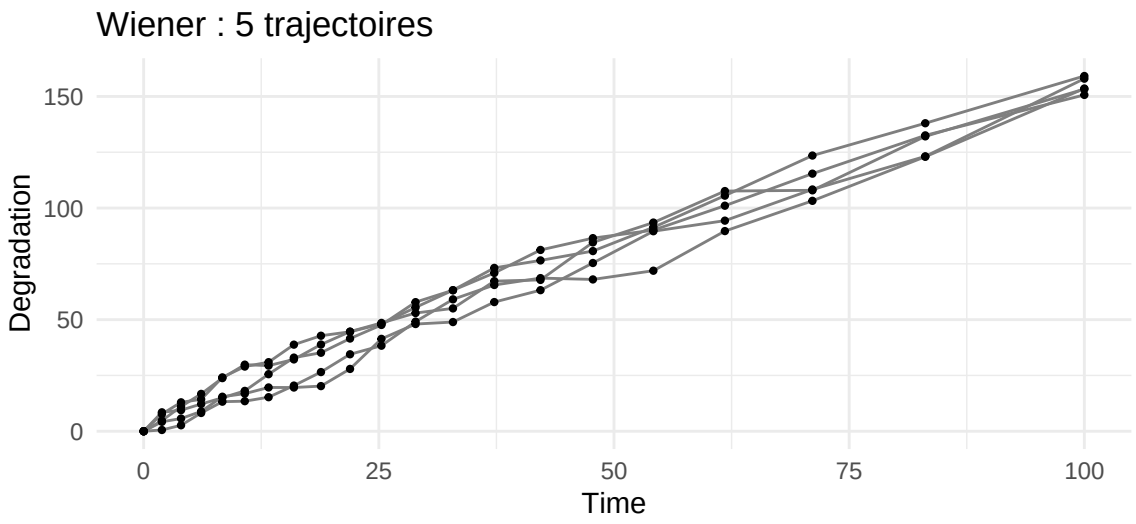
```
set.seed(132)
x <- sim_wiener(p3, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##              t        X1         X2        X3
## [1,]  0.000000  0.000000  0.0000000  0.000000
## [2,]  1.940035  4.230787  0.5691491  8.489047
## [3,]  3.974743  5.699667  2.6729907  9.454022
## [4,]  6.113840  8.879185  8.1968896 12.194740
## [5,]  8.368618 15.465729 13.2503463 14.987435
## [6,] 10.752309 16.842018 13.4664873 18.080960
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  4.231  1.469  3.180  6.587  1.376  2.773 -0.025  0.635  7.691 13.470
## [11]  6.643  0.893  8.950  5.330 12.177 14.230  4.739 13.966 14.832 34.817
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

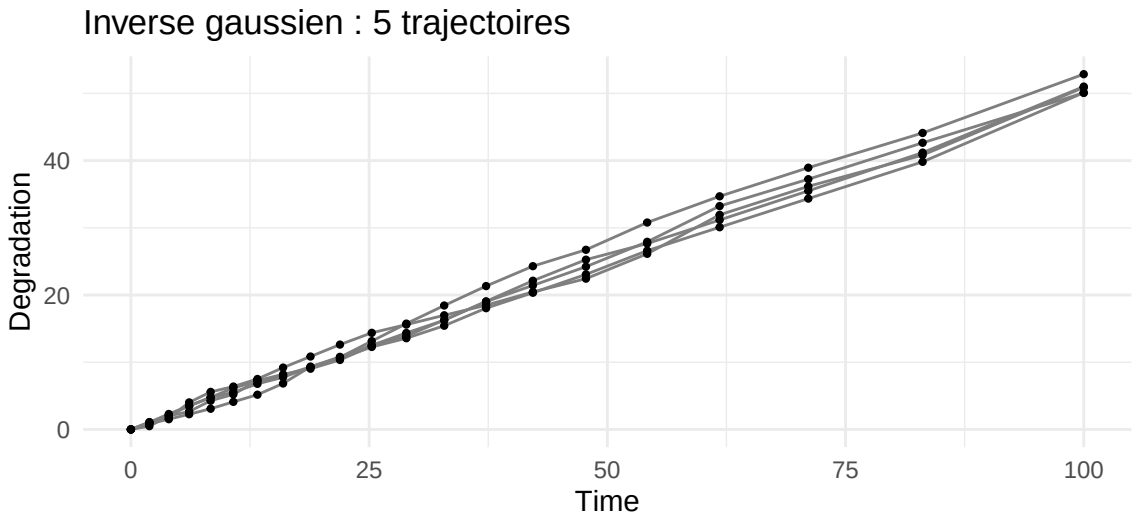
```
set.seed(133)
x <- sim_ig(p3, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1        X2       X3
## [1,]  0.000000 0.0000000 0.0000000 0.000000
## [2,]  1.940035 0.9390493 0.4884332 1.088656
## [3,]  3.974743 2.0880034 2.0127334 1.694220
## [4,]  6.113840 3.4899909 2.6631953 4.018758
## [5,]  8.368618 4.7335340 4.2683565 5.584057
## [6,] 10.752309 5.5338600 5.2210800 6.378180
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.939 1.149 1.402 1.244 0.800 1.245 0.987 1.482 1.473 2.417 2.603 2.697
## [13] 2.891 2.963 2.438 4.059 3.901 4.255 5.170 8.749
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

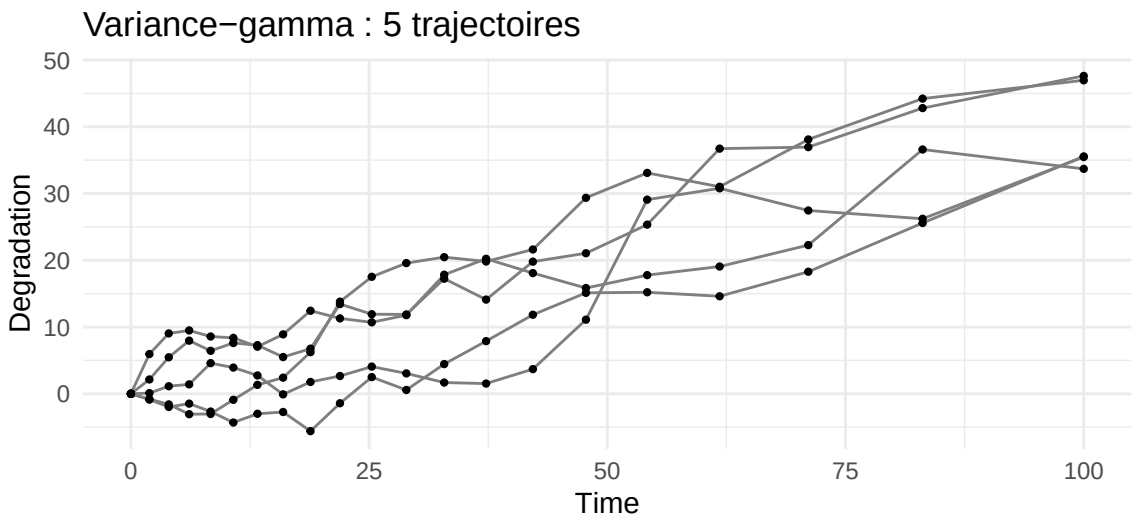
```
set.seed(134)
x <- sim_vg(p3, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t       X1        X2         X3
## [1,]  0.000000 0.000000 0.0000000  0.0000000
## [2,]  1.940035 5.953242 0.0991426 -0.8006998
## [3,]  3.974743 9.043546 1.1334917 -1.5900090
## [4,]  6.113840 9.487525 1.4108693 -3.0584259
## [5,]  8.368618 8.582190 4.5890451 -3.0247012
## [6,] 10.752309 8.374895 3.9255079 -0.8998503
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.953  3.090  0.444 -0.905 -0.207 -1.325  1.852  3.537 -1.146 -0.587
## [11]  1.073  6.059  2.383 -2.140 -2.239  1.929  1.291  3.208 14.324 -2.901
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

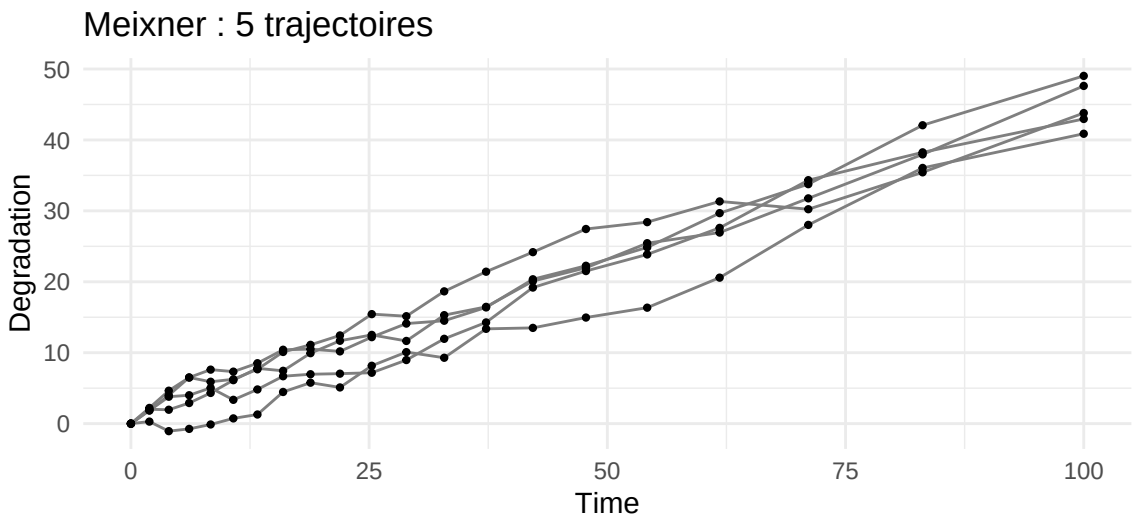
```
set.seed(135)
x <- sim_meixner(p3, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t         X1       X2       X3
## [1,]  0.000000  0.0000000 0.000000 0.000000
## [2,]  1.940035  0.2866694 2.024976 1.842790
## [3,]  3.974743 -1.0583616 1.958050 3.793688
## [4,]  6.113840 -0.7404516 2.908604 4.004078
## [5,]  8.368618 -0.1171312 4.326521 5.041161
## [6,] 10.752309  0.7393661 6.128973 3.356289
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  0.287 -1.345  0.318  0.623  0.856  0.548  3.176  1.307 -0.667  3.054
## [11]  1.925 -0.796  4.077  0.133  1.458  1.393  4.235  7.435  8.031  4.820
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

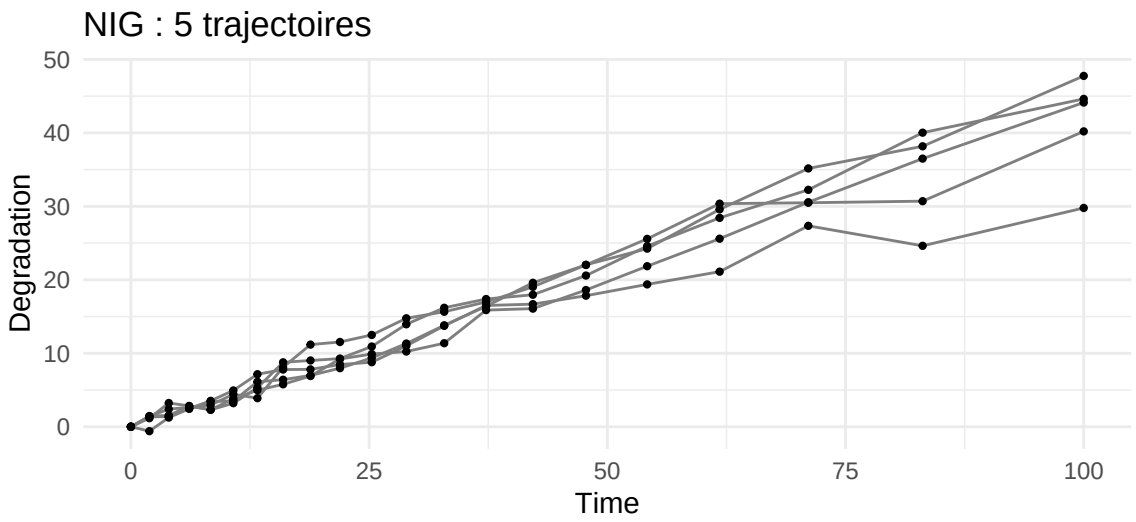
```
set.seed(136)
x <- sim_nig(p3, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##              t       X1       X2         X3
## [1,]  0.000000 0.000000 0.000000  0.0000000
## [2,]  1.940035 1.416675 1.163461 -0.5974616
## [3,]  3.974743 2.398329 3.233627  1.2486142
## [4,]  6.113840 2.639810 2.835734  2.4510788
## [5,]  8.368618 2.962988 2.302184  3.4433145
## [6,] 10.752309 4.418144 3.207621  3.5021879
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.417  0.982  0.241  0.323  1.455 -0.528  4.205  3.088  0.357  0.954
## [11]  2.269  0.891  1.326  2.062  2.999  2.201  5.348  5.570  3.014  9.580
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

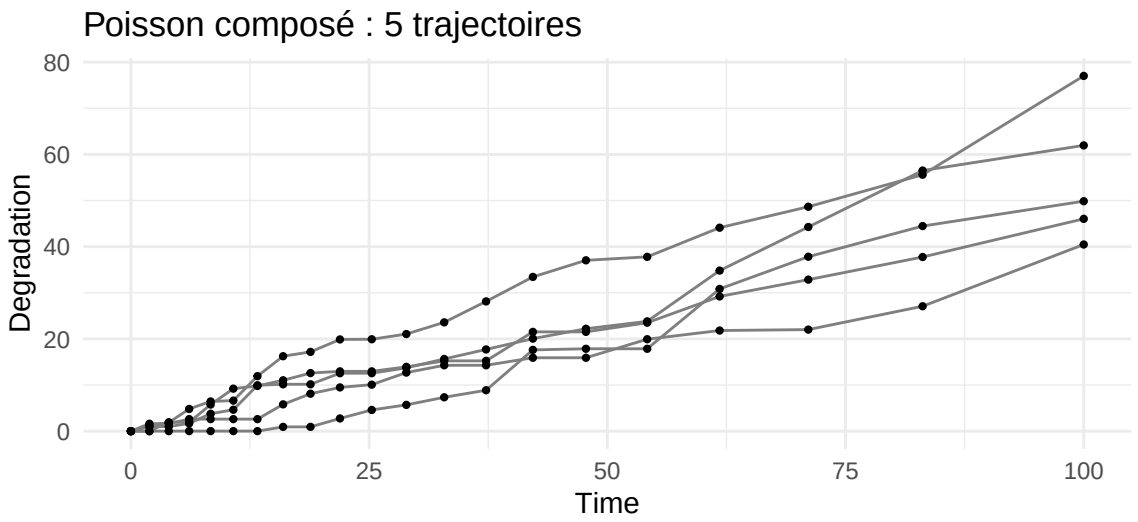
```
set.seed(137)
x <- sim_cpoisson(p3, lambda = 0.5)
head(x[, 1:4])
```

```
##              t       X1        X2       X3
## [1,]  0.000000 0.000000 0.0000000 0.000000
## [2,]  1.940035 1.126505 0.9923666 0.000000
## [3,]  3.974743 1.856240 0.9923666 1.922071
## [4,]  6.113840 4.813572 1.6737306 1.922071
## [5,]  8.368618 6.442241 3.7726864 5.759813
## [6,] 10.752309 6.619919 4.6405605 9.211759
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.127  0.730  2.957  1.629  0.178  5.304  4.321  0.932  2.706  0.037
## [11]  1.134  2.535  4.539  5.310  3.599  0.752  6.321  4.543  6.941 21.428
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```


Poisson composé : 5 trajectoires
Degradation
Time
0
20
40
60
80
0
25
50
75
100

Cas 4 : mixtes, beaucoup au début et à la fin

```
p4 <- inspection_times("inc-dec", n = 20, n_traj = 5, tmax = 100)
plot_plan(p4$time[p4$id == 1], "Beaucoup au début et à la fin, peu au milieu")
```



Modèle 1 : gamma

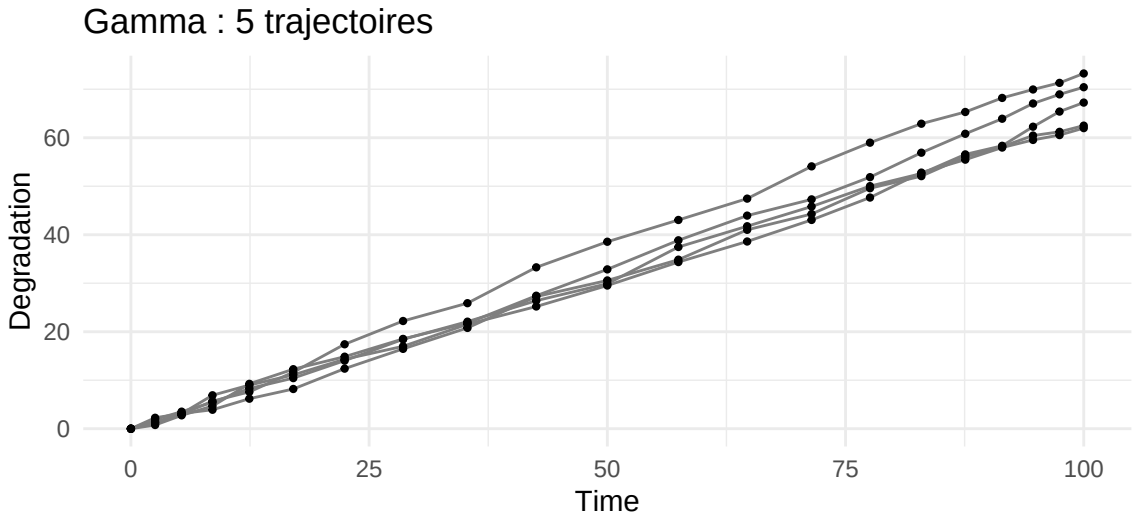
```
set.seed(141)
x <- sim_gamma(p4, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000  0.000000 0.000000  0.000000
## [2,]  2.540921  1.916327 1.174257  2.256682
## [3,]  5.322554  3.520553 3.137588  3.099718
## [4,]  8.562047  5.407939 3.909497  6.897871
## [5,] 12.431733  8.292020 6.188806  8.993691
## [6,] 17.042253 10.434968 8.191329 11.098034
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.916 1.604 1.887 2.884 2.143 3.602 4.399 3.640 4.331 3.499 7.567 4.281
## [13] 4.062 4.209 2.591 3.954 1.772 3.936 3.120 1.856
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

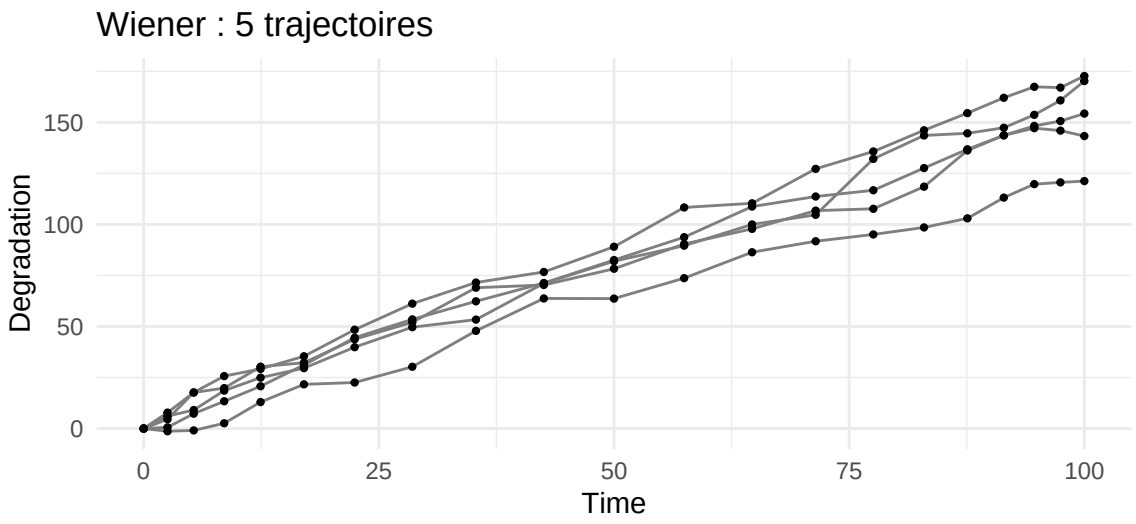
```
set.seed(142)
x <- sim_wiener(p4, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##              t        X1         X2        X3
## [1,]  0.000000  0.000000  0.0000000  0.000000
## [2,]  2.540921  7.811498 -1.3223131  4.565071
## [3,]  5.322554 17.668172 -0.9279949 17.670392
## [4,]  8.562047 25.696328  2.5898009 19.801467
## [5,] 12.431733 29.212882 12.9757735 30.310208
## [6,] 17.042253 35.402095 21.6501585 32.251385
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  7.811  9.857  8.028  3.517  6.189 13.028 12.753 10.340  5.187 12.348
## [11] 19.250  2.036 16.878  8.513 10.390  8.398  7.528  5.360 -0.391  5.692
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

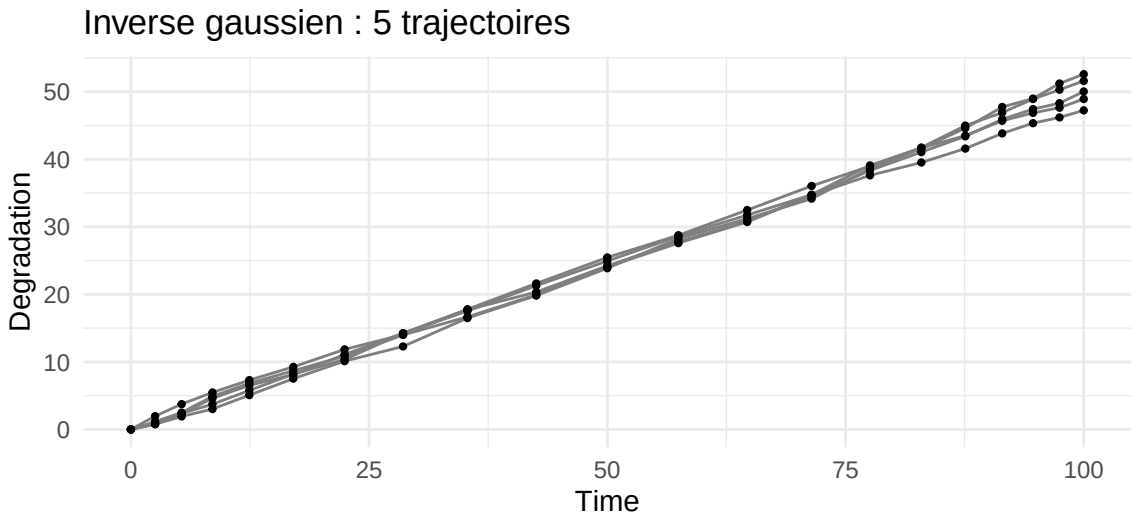
```
set.seed(143)
x <- sim_ig(p4, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000 0.0000000 0.000000 0.0000000
## [2,]  2.540921 0.7772242 1.956862 0.9506147
## [3,]  5.322554 1.9233357 3.745237 2.3784685
## [4,]  8.562047 3.0144983 5.472639 3.7276869
## [5,] 12.431733 5.0832297 7.300213 5.7749032
## [6,] 17.042253 7.5167033 9.259178 8.1877298
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.777 1.146 1.091 2.069 2.433 2.597 2.175 4.202 3.330 4.050 4.402 2.917
## [13] 2.977 4.438 2.984 3.016 3.141 1.239 2.242 1.378
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

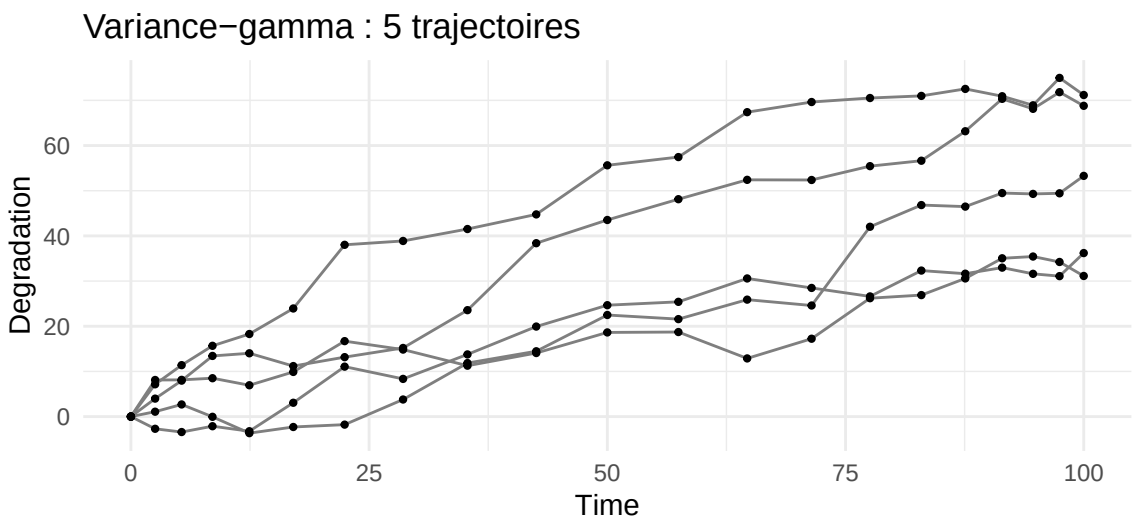
```
set.seed(144)
x <- sim_vg(p4, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t       X1        X2        X3
## [1,]  0.000000 0.000000  0.000000  0.000000
## [2,]  2.540921 8.101337  7.146385 -2.701680
## [3,]  5.322554 8.131089 11.386395 -3.421002
## [4,]  8.562047 8.493392 15.665319 -2.130697
## [5,] 12.431733 6.936565 18.279288 -3.241100
## [6,] 17.042253 9.887127 23.929366  3.061084
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  8.101  0.030  0.362 -1.557  2.951  6.814 -1.871 -3.565  2.809  4.555
## [11]  0.087 -5.843  4.361  8.967  0.692  3.686  4.459  0.392 -1.206 -3.099
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

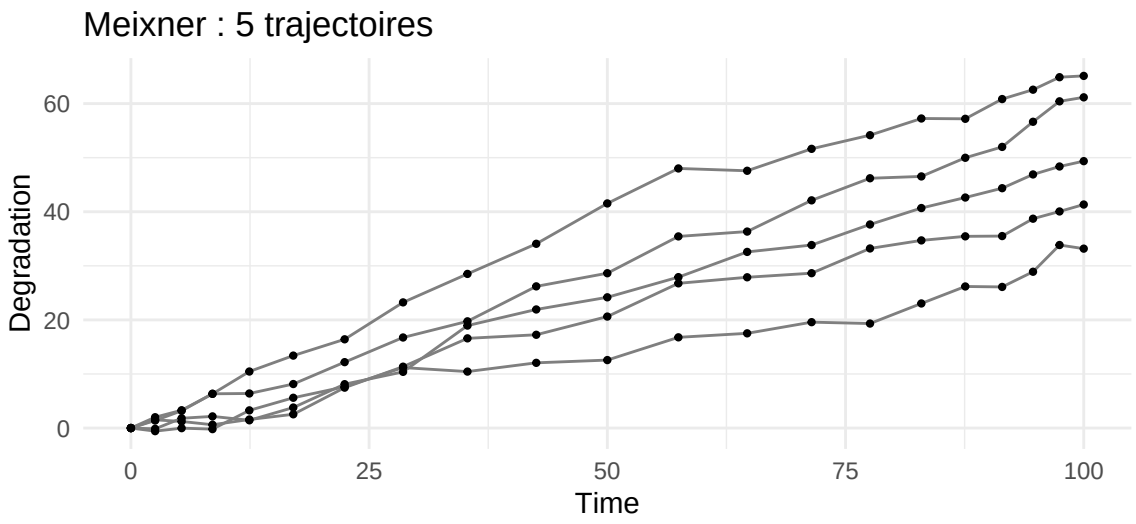
```
set.seed(145)
x <- sim_meixner(p4, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t        X1          X2        X3
## [1,]  0.000000  0.000000  0.00000000 0.0000000
## [2,]  2.540921  2.004889 -0.57081430 1.5262858
## [3,]  5.322554  3.288517 -0.01945328 1.2183414
## [4,]  8.562047  6.329637 -0.19546133 0.6123311
## [5,] 12.431733 10.456133  3.27044396 1.5840534
## [6,] 17.042253 13.399033  5.59940925 2.5455475
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.005  1.284  3.041  4.126  2.943  3.014  6.826  5.267  5.568  7.465
## [11]  6.469 -0.429  4.055  2.525  3.087 -0.058  3.650  1.735  2.309  0.241
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

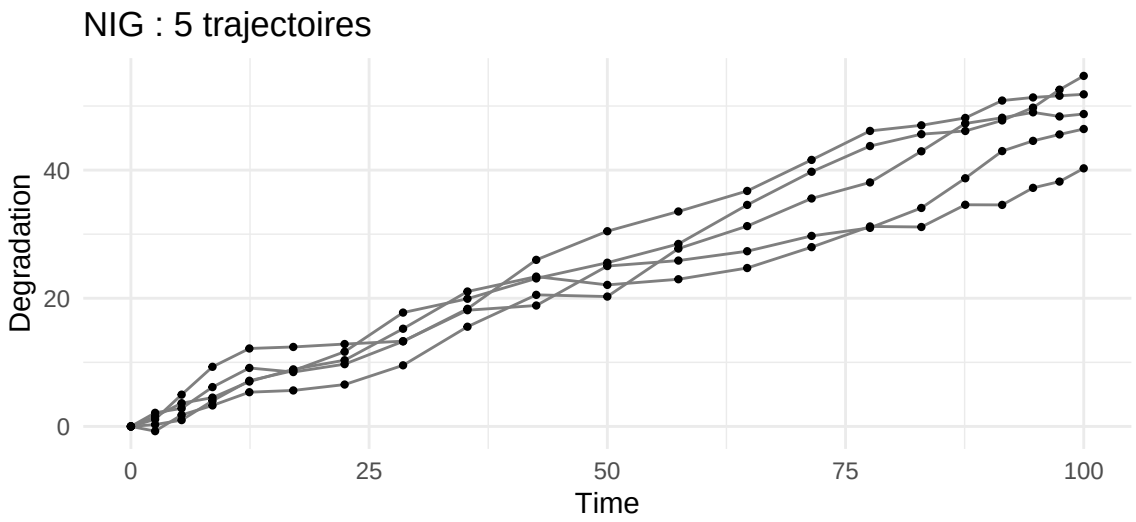
```
set.seed(146)
x <- sim_nig(p4, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##              t       X1         X2        X3
## [1,]  0.000000 0.000000  0.0000000 0.0000000
## [2,]  2.540921 1.561046 -0.7367223 0.2876020
## [3,]  5.322554 3.624618  1.7851754 0.9848451
## [4,]  8.562047 4.495267  3.2748986 3.9989290
## [5,] 12.431733 7.001454  5.3411814 7.1301402
## [6,] 17.042253 8.906618  5.6123716 8.7004438
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.561  2.064  0.871  2.506  1.905  1.425  4.912  5.809  2.327 -1.294
## [11]  0.883  1.751  3.257  3.229 -0.084  3.469 -0.022  2.659  0.981  2.066
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

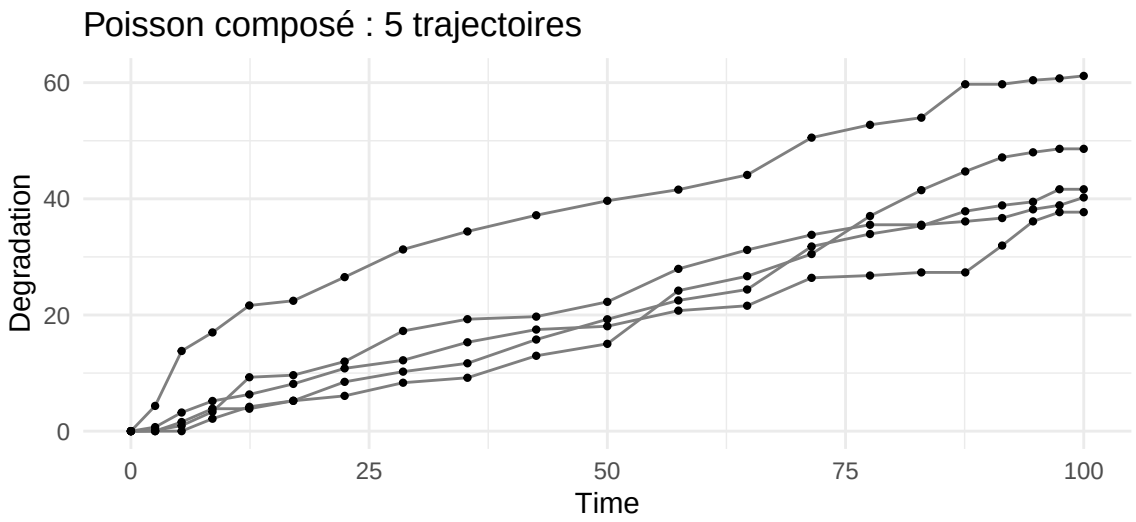
```
set.seed(147)
x <- sim_cpoisson(p4, lambda = 0.5)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000 0.0000000  0.000000 0.0000000
## [2,]  2.540921 0.1210714  4.345207 0.6949144
## [3,]  5.322554 0.9519051 13.802905 3.2070645
## [4,]  8.562047 3.3447454 16.996240 5.1935449
## [5,] 12.431733 9.2809750 21.638766 6.3193773
## [6,] 17.042253 9.6390935 22.445798 8.1355463
```

```
round(diff(x[, "X1"]), 3)
```

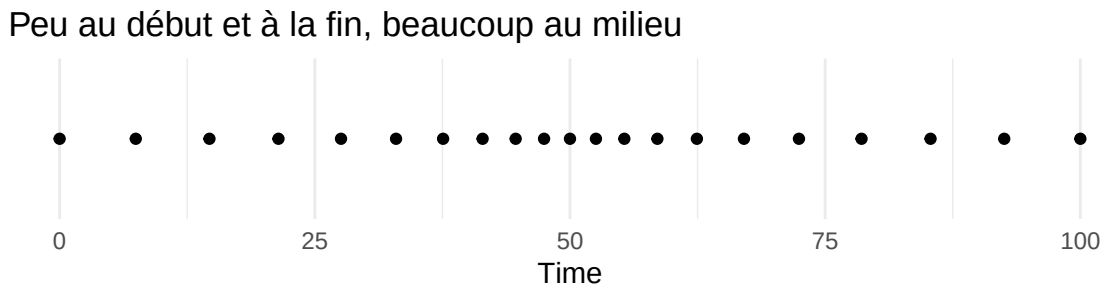
```
##  [1] 0.121 0.831 2.393 5.936 0.358 2.346 5.264 2.027 0.440 2.544 5.680 3.261
## [13] 2.598 1.733 0.000 0.576 0.580 1.499 0.698 1.351
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```


Poisson composé : 5 trajectoires
Degradation
Time
0
20
40
60
0
25
50
75
100

Cas 5 : mixtes, beaucoup au milieu

```
p5 <- inspection_times("dec-inc", n = 20, n_traj = 5, tmax = 100)
plot_plan(p5$time[p5$id == 1], "Peu au début et à la fin, beaucoup au milieu")
```



Modèle 1 : gamma

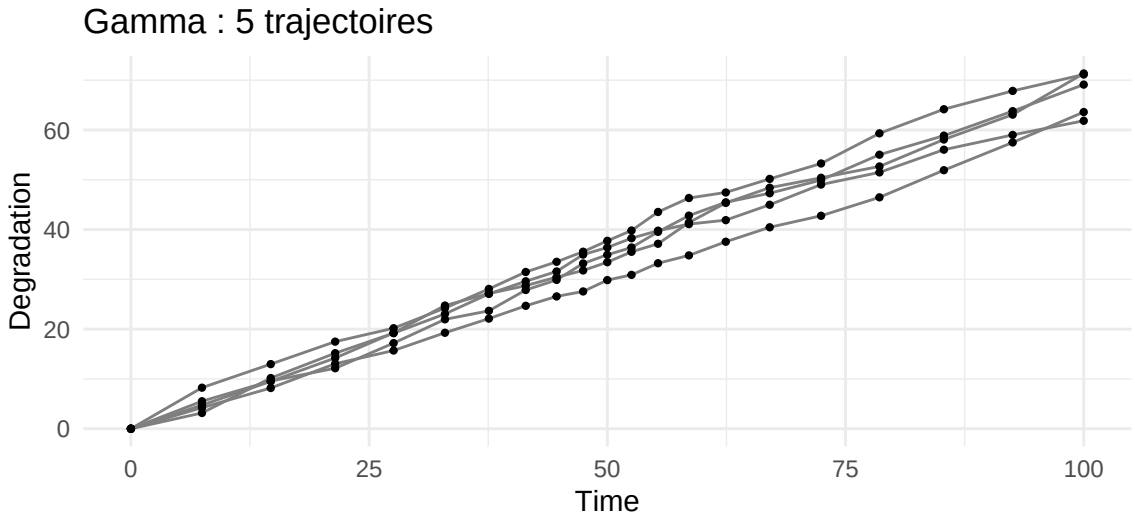
```
set.seed(151)
x <- sim_gamma(p5, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  7.459079  4.741041  5.521434  4.228339
## [3,] 14.677446  9.508096  9.575761  8.175986
## [4,] 21.437953 14.203284 12.146490 13.038912
## [5,] 27.568267 19.244251 17.192410 15.703649
## [6,] 32.957747 24.736728 21.962976 19.268742
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 4.741 4.767 4.695 5.041 5.492 2.407 1.586 1.710 1.344 1.661 2.085 1.614
## [13] 4.256 3.951 3.052 1.993 2.272 5.417 4.993 8.263
```

```
plot_traj(x, "Gamma : 5 trajectoires")
```



Modèle 2 : Wiener

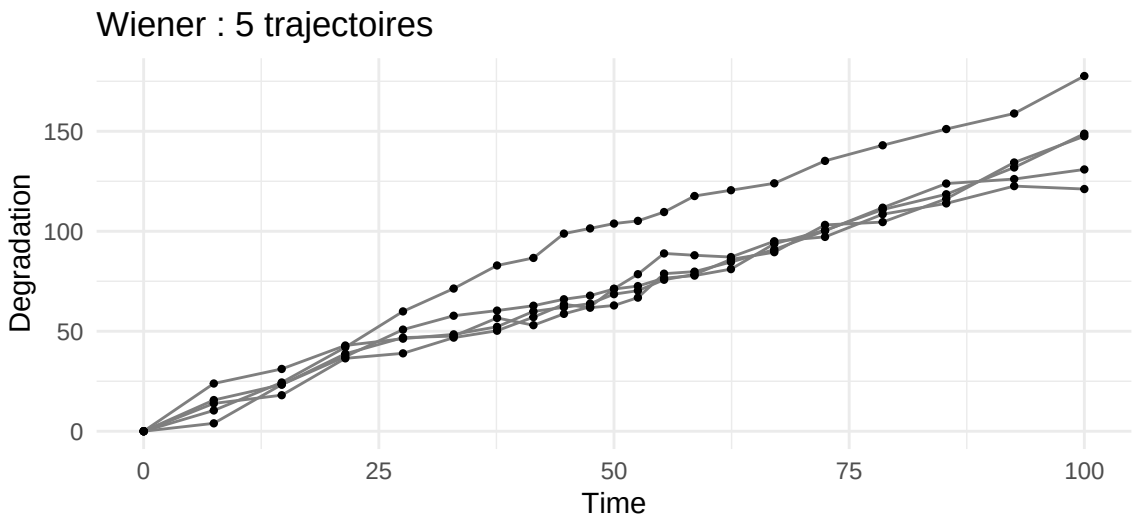
```
set.seed(152)
x <- sim_wiener(p5, mu = 1.5, sigma = 2)
head(x[, 1:4])
```

```
##              t       X1       X2       X3
## [1,]  0.000000  0.00000  0.00000  0.000000
## [2,]  7.459079 23.86975 15.59424  3.958588
## [3,] 14.677446 31.14916 23.32680 23.360176
## [4,] 21.437953 42.87639 38.71552 37.302685
## [5,] 27.568267 46.29445 46.82211 50.815925
## [6,] 32.957747 48.44725 47.57995 57.756556
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 23.870  7.279 11.727  3.418  2.153  3.777  7.718  2.015  1.883  4.704
## [11]  1.753  5.451  2.693  7.475  3.619 13.646  1.412 11.651 18.147 13.120
```

```
plot_traj(x, "Wiener : 5 trajectoires")
```



Modèle 3 : inverse gaussien

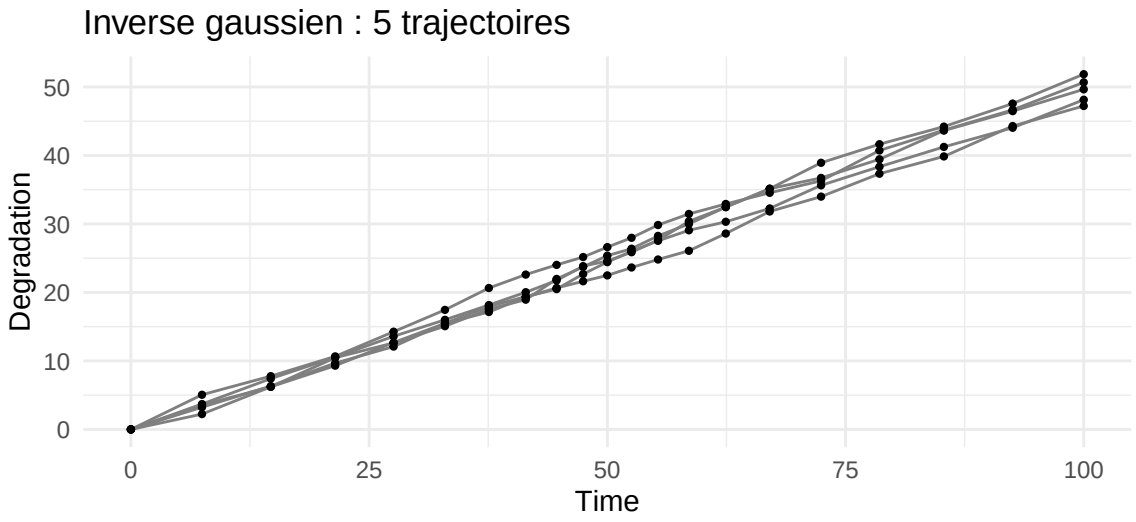
```
set.seed(153)
x <- sim_ig(p5, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  7.459079  2.238359  3.240561  3.557905
## [3,] 14.677446  6.217203  6.325747  6.219868
## [4,] 21.437953 10.468404  9.677937  9.307220
## [5,] 27.568267 12.628815 12.094830 12.680646
## [6,] 32.957747 15.074764 15.487383 15.478020
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 2.238 3.979 4.251 2.160 2.446 2.483 1.386 3.026 1.761 1.638 1.012 1.874
## [13] 1.749 2.454 2.684 3.788 2.710 2.572 3.362 4.294
```

```
plot_traj(x, "Inverse gaussien : 5 trajectoires")
```



Modèle 4 : variance-gamma

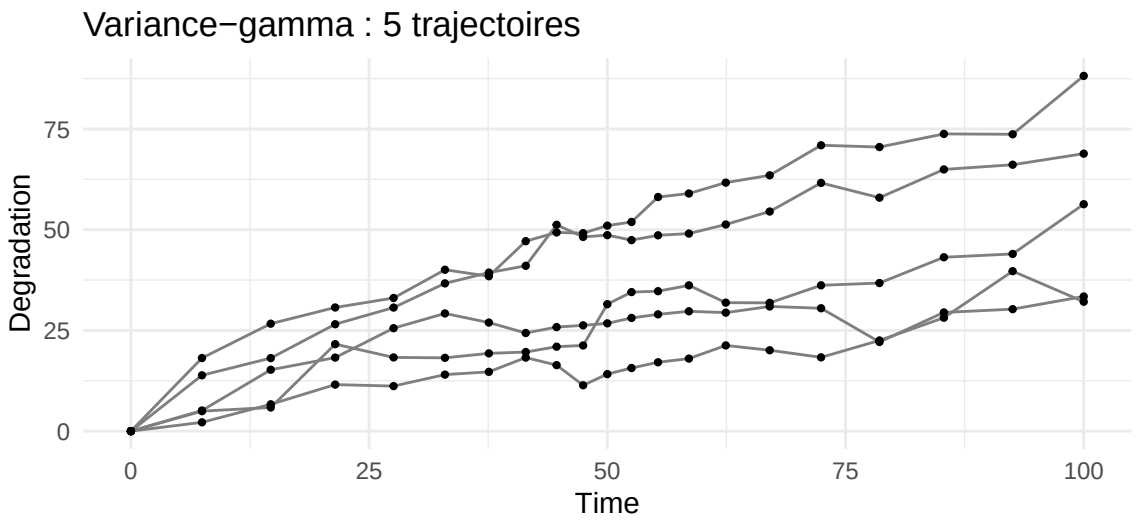
```
set.seed(154)
x <- sim_vg(p5, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t        X1        X2       X3
## [1,]  0.000000  0.000000  0.000000  0.00000
## [2,]  7.459079  2.190125  4.960023 13.85976
## [3,] 14.677446  6.667534  5.870367 18.13231
## [4,] 21.437953 11.550031 21.591089 26.53618
## [5,] 27.568267 11.182291 18.299595 30.67430
## [6,] 32.957747 14.030616 18.197226 36.67401
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  2.190  4.477  4.882 -0.368  2.848  0.685  3.542 -1.873 -4.993  2.766
## [11]  1.512  1.428  0.910  3.264 -1.194 -1.765  4.196  5.645 11.551 -7.583
```

```
plot_traj(x, "Variance-gamma : 5 trajectoires")
```



Modèle 5 : Meixner

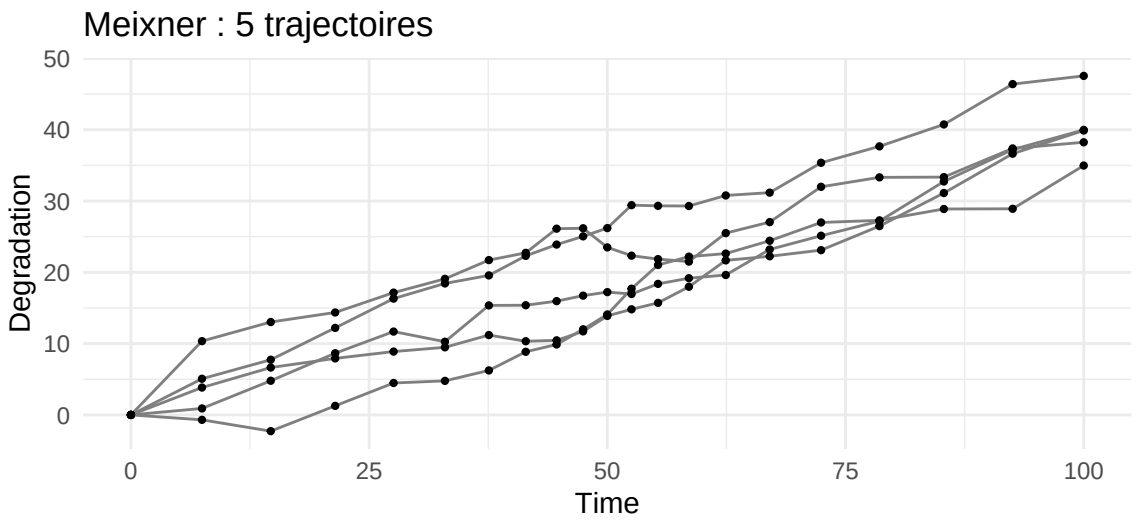
```
set.seed(155)
x <- sim_meixner(p5, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t        X1       X2       X3
## [1,]  0.000000  0.000000 0.000000  0.00000
## [2,]  7.459079  5.071196 3.839912 10.35035
## [3,] 14.677446  7.739227 6.643030 13.02948
## [4,] 21.437953 12.206749 7.927603 14.36807
## [5,] 27.568267 16.315425 8.886013 17.15169
## [6,] 32.957747 18.441474 9.482853 19.08732
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  5.071  2.668  4.468  4.109  2.126  1.128  2.741  1.584  1.152  1.161
## [11]  3.216 -0.093 -0.024  1.478  0.398  4.188  2.306  3.073  5.657  1.151
```

```
plot_traj(x, "Meixner : 5 trajectoires")
```



Modèle 6 : normal inverse gaussien (NIG)

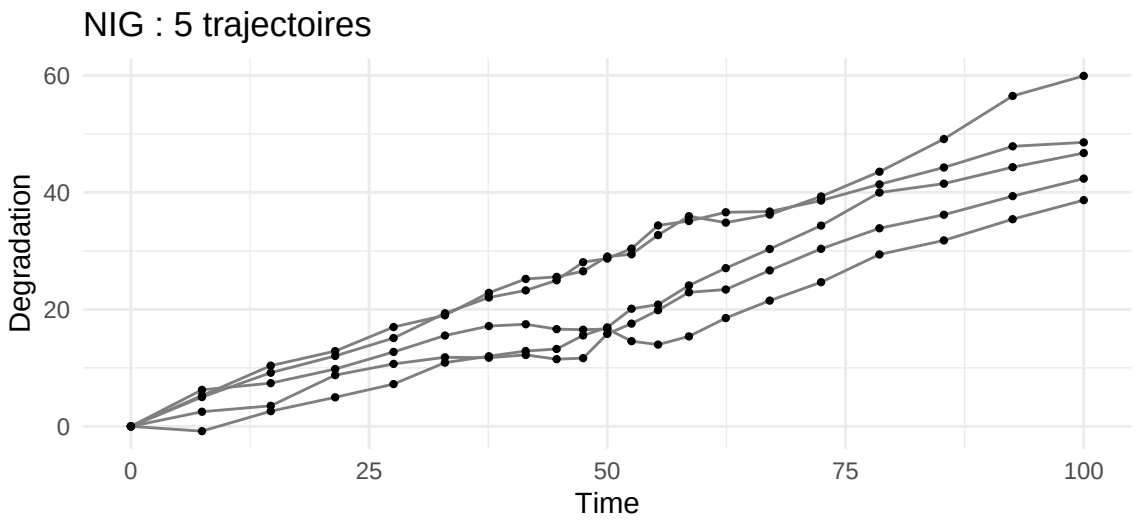
```
set.seed(156)
x <- sim_nig(p5, alpha = 2, beta = 0.5, delta = 1, mu = 0.2)
head(x[, 1:4])
```

```
##              t         X1        X2        X3
## [1,]  0.000000  0.0000000  0.000000  0.000000
## [2,]  7.459079 -0.8154205  6.241988  5.009794
## [3,] 14.677446  2.6002278  7.387626  9.169580
## [4,] 21.437953  4.9651138  9.796440 12.038473
## [5,] 27.568267  7.2308250 12.724568 15.099153
## [6,] 32.957747 10.8952491 15.527961 19.331089
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] -0.815  3.416  2.365  2.266  3.664  1.096  0.888  0.346  2.322  1.370
## [11]  3.189  0.709  3.267  2.964  3.271  4.003  5.652  1.520  2.809  2.423
```

```
plot_traj(x, "NIG : 5 trajectoires")
```



Modèle 7 : Poisson composé

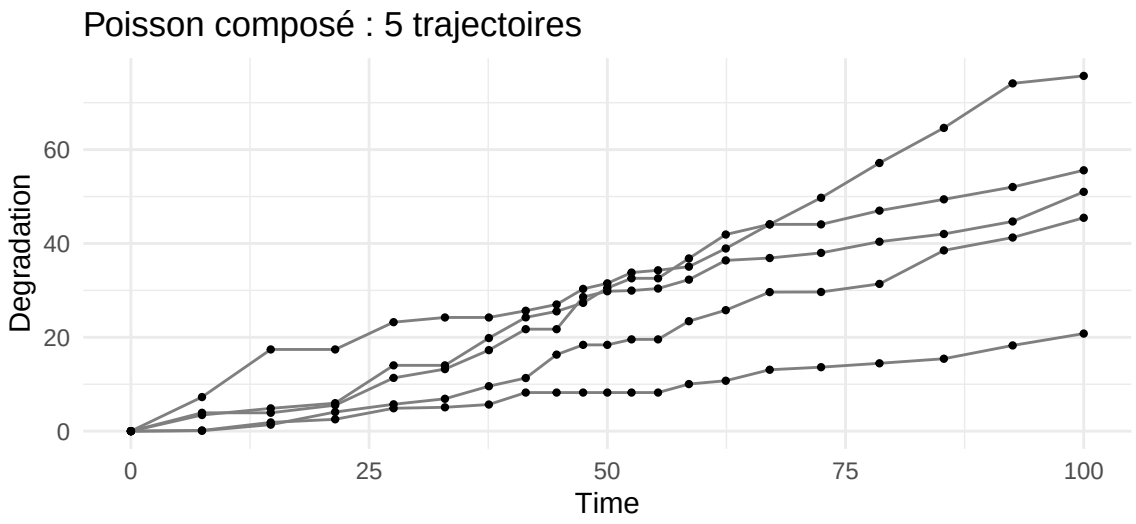
```
set.seed(157)
x <- sim_cpoisson(p5, lambda = 0.5)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000 0.0000000
## [2,]  7.459079  3.911296  3.438826 0.1384595
## [3,] 14.677446  3.911296  4.852867 1.8628205
## [4,] 21.437953  5.528340  5.961461 2.5230237
## [5,] 27.568267 11.344923 14.017337 4.8769255
## [6,] 32.957747 13.224447 14.017337 5.0783270
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 3.911 0.000 1.617 5.817 1.880 4.040 4.465 0.000 6.858 1.222 0.144 0.439
## [13] 1.896 4.099 0.505 1.094 2.385 1.647 2.663 6.315
```

```
plot_traj(x, "Poisson composé : 5 trajectoires")
```



ewpage

III - Processus de Lévy à échelle de temps

Une fonction d'échelle de temps A (fonction croissante, continue à droite avec des limites à gauche, $A(0) = 0$ et $A(t) \to +\infty$) donne le processus de Lévy à échelle de temps $X_t = Y_{A(t)}$, où Y est un processus de Lévy. Le paramètre que A remplace (identifiabilité) n'est alors pas donné :

Processus	Fonction	$X_t - X_s$, $\Delta A = A(t) - A(s)$	Avec $A(t) = at$ ou μt
Gamma	`sim_gamma(t, b, A)`	$\Gamma(\Delta A, b)$	gamma de forme a
Wiener	`sim_wiener(t, sigma, A)`	$\mathcal{N}(\Delta A, \sigma^2 \Delta A)$	Wiener $(\mu, \sigma\sqrt{\mu})$
Inverse gaussien	`sim_ig(t, eta, A)`	$IG(\Delta A, \eta \Delta A^2)$	IG $(\mu, \eta\mu^2)$

Pour le Wiener, $X_t = \mu t + \sigma B_{\mu t}$ a la loi de $\mu t + \sigma\sqrt{\mu}\, B_t$, car $(B_{\mu t})_{t\geq 0}$ a la loi de $(\sqrt{\mu}\, B_t)_{t\geq 0}$.

On prend $A(t) = 0.02\, t^{1.5}$, de sorte que $A(100) = 20$, et les 5 trajectoires du cas 2 de la partie II (pas décroissants).

Modèle 1 : gamma à échelle de temps (Cha et Mercier, 2019, exemple 2)

```
A <- function(t) 0.02 * t^1.5
set.seed(301)
x <- sim_gamma(p2, b = 3, A = A)
x
```

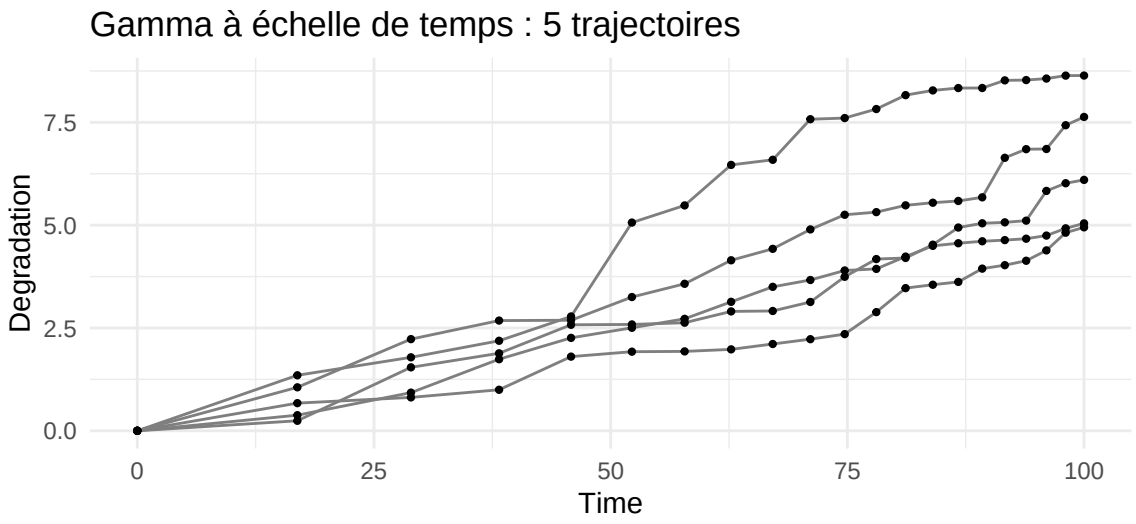
```
## 5 simulated trajectory(ies) of the time-scaled gamma process, same inspection times (matrix)
## parameters: b = 3
## time-scaling function: A = function (t)  0.02 * t^1.5
##
##             t        X1       X2        X3        X4       X5
## [1,]  0.00000 0.0000000 0.000000 0.0000000 0.0000000 0.000000
```

```
## [2,] 16.90921 0.3792614 1.349112 0.2429382 0.6721473 1.057136
## [3,] 28.90648 0.9275601 1.786393 1.5432021 0.8117168 2.228763
## [4,] 38.21229 1.7404953 2.188722 1.8841561 0.9973612 2.680303
## [5,] 45.81569 2.2589926 2.780103 2.5772813 1.8032195 2.687744
## [6,] 52.24427 2.5044998 5.063078 2.5860069 1.9234865 3.252174
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 0.379 0.548 0.813 0.518 0.246 0.218 0.413 0.364 0.166 0.231 0.039 0.299
## [13] 0.261 0.064 0.048 0.028 0.034 0.076 0.176 0.120
```

```
plot_traj(x, "Gamma à échelle de temps : 5 trajectoires")
```



Modèle 2 : Wiener à échelle de temps (Cha et Mercier, 2019, exemple 1)

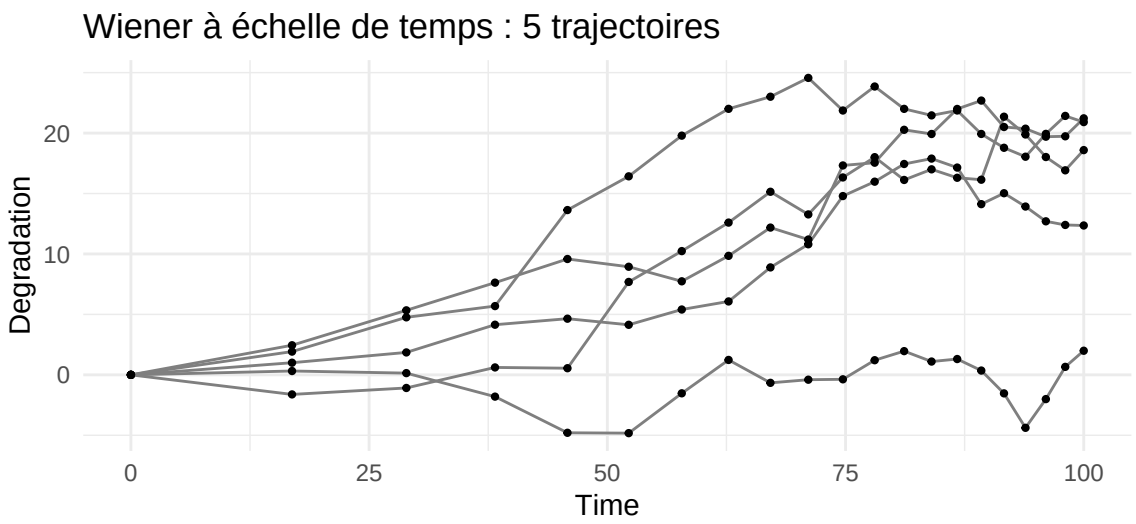
```
set.seed(302)
x <- sim_wiener(p2, sigma = 2, A = A)
head(x[, 1:4])
```

```
##             t        X1       X2         X3
## [1,]  0.00000  0.000000 0.000000  0.0000000
## [2,] 16.90921  1.923276 2.448910  0.3174737
## [3,] 28.90648  4.757541 5.337586  0.1435598
## [4,] 38.21229  5.687673 7.626493 -1.8019695
## [5,] 45.81569 13.631897 9.583937 -4.7940957
## [6,] 52.24427 16.422778 8.941656 -4.8223706
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1]  1.923  2.834  0.930  7.944  2.791  3.365  2.218  1.002  1.557 -2.697
## [11]  1.984 -1.849 -0.540  0.405 -1.941 -1.144 -0.739  1.888  1.488 -0.512
```

```
plot_traj(x, "Wiener à échelle de temps : 5 trajectoires")
```



Modèle 3 : inverse gaussien à échelle de temps (Cha et Mercier, 2019, exemple 3)

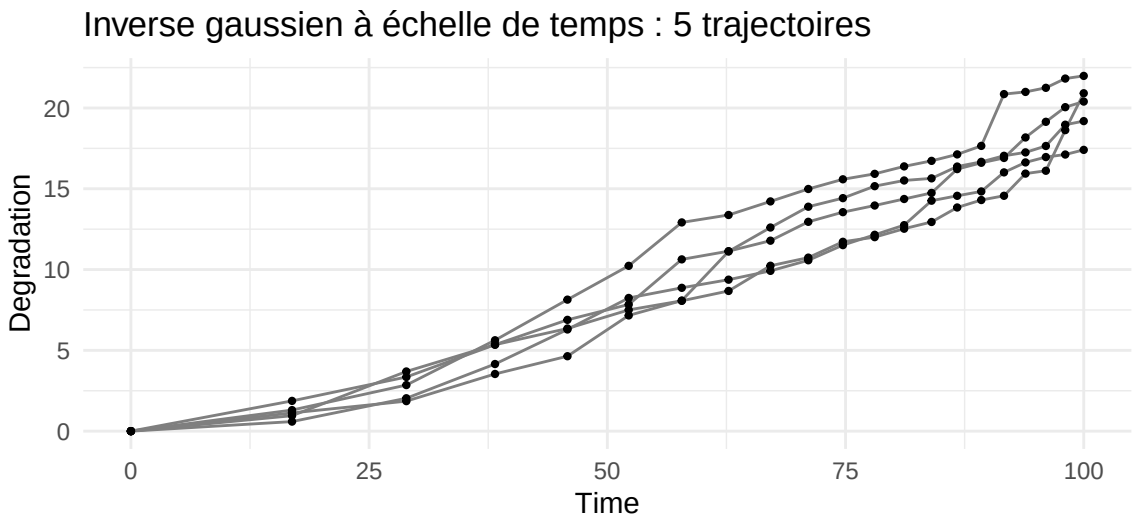
```
set.seed(303)
x <- sim_ig(p2, eta = 2, A = A)
head(x[, 1:4])
```

```
##             t       X1        X2       X3
## [1,]  0.00000 0.000000 0.0000000 0.000000
## [2,] 16.90921 1.871329 0.9560403 1.128459
## [3,] 28.90648 3.349287 3.6921305 1.856956
## [4,] 38.21229 5.349771 5.3504349 3.538414
## [5,] 45.81569 6.887926 6.3527608 4.637236
## [6,] 52.24427 7.839307 7.5044765 7.160367
```

```
round(diff(x[, "X1"]), 3)
```

```
##  [1] 1.871 1.478 2.000 1.538 0.951 2.791 0.508 0.648 1.168 0.597 0.412 0.403
## [13] 0.375 1.475 0.390 0.299 1.268 0.969 0.905 0.348
```

```
plot_traj(x, "Inverse gaussien à échelle de temps : 5 trajectoires")
```



Vérification : avec $A(t) = at$, on retrouve le processus gamma de forme a

Avec la même graine, le gamma à échelle de temps avec $A(t) = 2t$ et le gamma $(a, b) = (2, 3)$ donnent les mêmes trajectoires :

```
set.seed(304)
y1 <- sim_gamma(p2, b = 3, A = function(t) 2 * t)
set.seed(304)
y2 <- sim_gamma(p2, a = 2, b = 3)
all.equal(unclass(y1)[, -1], unclass(y2)[, -1], check.attributes = FALSE)
```

```
## [1] TRUE
```